DISTANCE LEARNING CENTRE

AHMADU BELLO UNIVERSITY

ZARIA, NIGERIA

COURSE MATERIAL

FOR

Course Code &Title: LIBS 897 Preservation Of Records

Programme Title: MASTER IN INFORMATION MANAGEMENT

ACKNOWLEDGEMENT

We acknowledge the use of the Courseware of the National Open University of Nigeria (NOUN) as the primary resource. Internal reviewers in the Ahmadu Bello University have also been duly listed.

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COURSE STUDY GUIDE

i. COURSE INFORMATION

Course Code: LIBS 897

Course Title: Preservation of Records.

Credit Units: 3

Semester: One

ii. COURSE INTRODUCTION AND DESCRIPTION

Introduction:

You are welcome to LIBS 897, I will be your guide in the span of this course, you should feel free to ask questions when in a form of difficulties relating this course, and my door is always open.

Preservation of records is an old area in information management. It originated from the time of the creation of mankind when only human brain was used to store information for its preservation. With the invention of writing, people used different local writing surfaces to record and preserve their information. One of those surfaces was clay, which was popularly called **clay table**. This material was used in Mesopotamia, where people recorded their activities (farming, education, buying and selling, wedding ceremony etc.) about 5000 BC. Another local writing surface was the Papyrus. This was used in Egypt by the people in order to record their activities. This was done around 3000BC. Furthermore, animal skins, bones, Pergamum and Parchment were used as local writing surfaces. Others were leaves, bark of tree, wax etc.All these early local writing surfaces were used to record the activities of the people and they were preserved for their use. However, with the invention of paper, it was used for records creation, which was preserved for daily information use. After the invention of the paper, electronic devices were invented,

which were used for records preservation. Computer was an example of the electronic devices used in preserving records up to today. Though, preservation of records is necessary for human daily activities (farming, education, healthcare etc.),early people preserved their records for their activities. Today, people preserve their records in order to be able to know what happened before, what is happening today so as to plan for the future in their daily activities. Organisations (schools, hospitals, companies, banks etc.), generate and receive records, both paper based and electronic, in the course of their daily activities (meetings, reporting, supervision etc.). As a result of that, such records, need to be preserved in order to provide the information which will guide the activities they carry out. Also, individuals generate and receive records in the course of their interactions (buying, selling, visitation etc.) with others. They need such records because of the information they contain. As such, preservation of their records is an important issue.

Description:

The course is focused on the preservation of records (current and archives) regardless of their formats. Preservation of records in information Centre's such as libraries, archives, museums, administrative offices among others. Preservation of paper- based records was discussed. **Digital preservation** of records was described. This is very important as digital technology is found everywhere. Strategies of digital preservation of records such as migration, digitization, and emulation among others were described. Importance of preservation of digital records, maintenance of **digital records** and requirements for the preservation of digital records such as human being, facilities, equipment, knowledge etc. were described.

iii. COURSE PREREQUISITES

You should note that although this course has no subject pre-requisite, you are expected to have:

1. Satisfactory level of English proficiency
2. Basic Computer Operations proficiency
3. Online interaction proficiency
4. Web 2.0 and Social media interactive skills

iv. COURSE LEARNING RESOURCES

You should know that, there are no compulsory textbooks for the course. Notwithstanding, you are encouraged to consult some of those listed for further reading at the end of each study session.

i. Course Textbooks

Ancientpage.com. (2017). Cuneiform Tablets: One of the Earliest Systems of Writing Invented By the Sumerians. Retrieved from http://www.ancientpages.com/2017/05/08/cuneiform-tablets-one-of-the-earliest-systems-of-writing-invented-by-the-sumerians/.Accessed on 27/12/2018.

Regents of the University of Michigan.(2018). Ancient Writing Materials. Retrieved from https://www.lib.umich.edu/papyrology-collection/ancient-writing-materials.Accessed on 27/12/2018.

v. COURSE OUTCOMES

After studying this course, you should be able to:

1. Apply the knowledge acquired from the course to preservation of records irrespective of the format and environment.
2. Identify issues in preservation of records and give them professional treatment in order to enhance the situation.
3. Analyse cases in preservation of records and proffer solution when the need arises.

vi. ACTIVITIES TO MEET COURSE OBJECTIVES

The distance learning system of education is quite different from the traditional University system. Here, the study sessions replace the University lecturers, thus conferring a unique advantage on you. For instance, you can read and work through specially designed study materials at your own pace and at a time and place that suit you best.

You should understand right from the onset that the contents of the course are to be worked at and understood step by step and not to be read like a novel. The best way is to read a study session quickly in order to see the general run of the contents and then to re-read it carefully, making sure that the contents are understood step by step.

You should be prepared at this stage to spend a much longer quality time on some study sessions that may appear difficult. A paper and pen may be necessary. Ensure that you make necessary notes and summaries where necessary for future reference.

Moreover, the key to success in this course is to study each topic slowly in order to ensure full apprehension before moving on to the next one. In the event any topic is not fully understood, there is every need to go through it over and over again by

reworking illustrations in the topic. This is based on the fact that practice plays significant role in understanding subjects that deal with highly technical issues. Also, there will be series of group and individual assignments that you are expected to do and submit within the defined time limit which will serve as part of your assessment.

Final, proviso of e-tutor's e-mail and telephone number will be made to enable you seek clarification on things that are not cleared to you.

Specifically, this course shall comprise of the following activities:

1. Studying courseware
2. Listening to course audios
3. Watching relevant course videos
4. Field activities, industrial attachment or internship, laboratory or studio work (whichever is applicable)
5. Course assignments (individual and group)
6. Forum discussion participation
7. Tutorials (optional)
8. Semester examinations (CBT and essay based).

vii. TIME (TO COMPLETE SYLABUS/COURSE)

To cope with this course, you would be expected to commit a minimum of 3 hours Weekly for the Course.

viii. GRADING CRITERIA AND SCALE

Grading Criteria

A. Formative assessment

Grades will be based on the following:

Individual assignments/test (CA 1,2etc.)	20
Group assignments (GCA 1, 2 etc.)	10
Discussions/Quizzes/Out of class engagements etc	10

B. Summative assessment (Semester examination)

CBT based	30
Essay based	30
TOTAL	**100%**

C. Grading Scale :

A = 70-100

B = 60 – 69

C = 50 - 59

D = 45-49

F = 0-44

D. Feedback

Courseware based:

1. In-text questions and answers (answers preceding references)
2. Self-assessment questions and answers (answers preceding references)

Tutor based:

1. Discussion Forum tutor input
2. Graded Continuous assessments

Student based:

1. Online programme assessment (administration, learning resource, deployment, and assessment).

ix. LINKS TO OPEN EDUCATION RESOURCES

OSS Watch provides tips for selecting open source, or for procuring free or open software.

SchoolForge and SourceForge are good places to find, create, and publish open software. SourceForge, for one, has millions of downloads each day.

Open Source Education Foundation and Open Source Initiative, and other organisation like these, help disseminate knowledge.

Creative Commons has a number of open projects from Khan Academy to Curriki where teachers and parents can find educational materials for children or learn about Creative Commons licenses. Also, they recently launched the School of Open that offers courses on the meaning, application, and impact of "openness."

Numerous open or open educational resource databases and search engines exist. Some examples include:

- OEDb: over 10,000 free courses from universities as well as reviews of colleges and rankings of college degree programmes
- Open Tapestry: over 100,000 open licensed online learning resources for an academic and general audience
- OER Commons: over 40,000 open educational resources from elementary school through to higher education; many of the elementary, middle, and high school resources are aligned to the Common Core State Standards
- Open Content: a blog, definition, and game of open source as well as a friendly search engine for open educational resources from MIT, Stanford, and other universities with subject and description listings

- Academic Earth: over 1,500 video lectures from MIT, Stanford, Berkeley, Harvard, Princeton, and Yale
- JISC: Joint Information Systems Committee works on behalf of UK higher education and is involved in many open resources and open projects including digitising British newspapers from 1620-1900!

Other sources for open education resources

Universities

- The University of Cambridge's guide on Open Educational Resources for Teacher Education (ORBIT)
- OpenLearn from Open University in the UK

Global

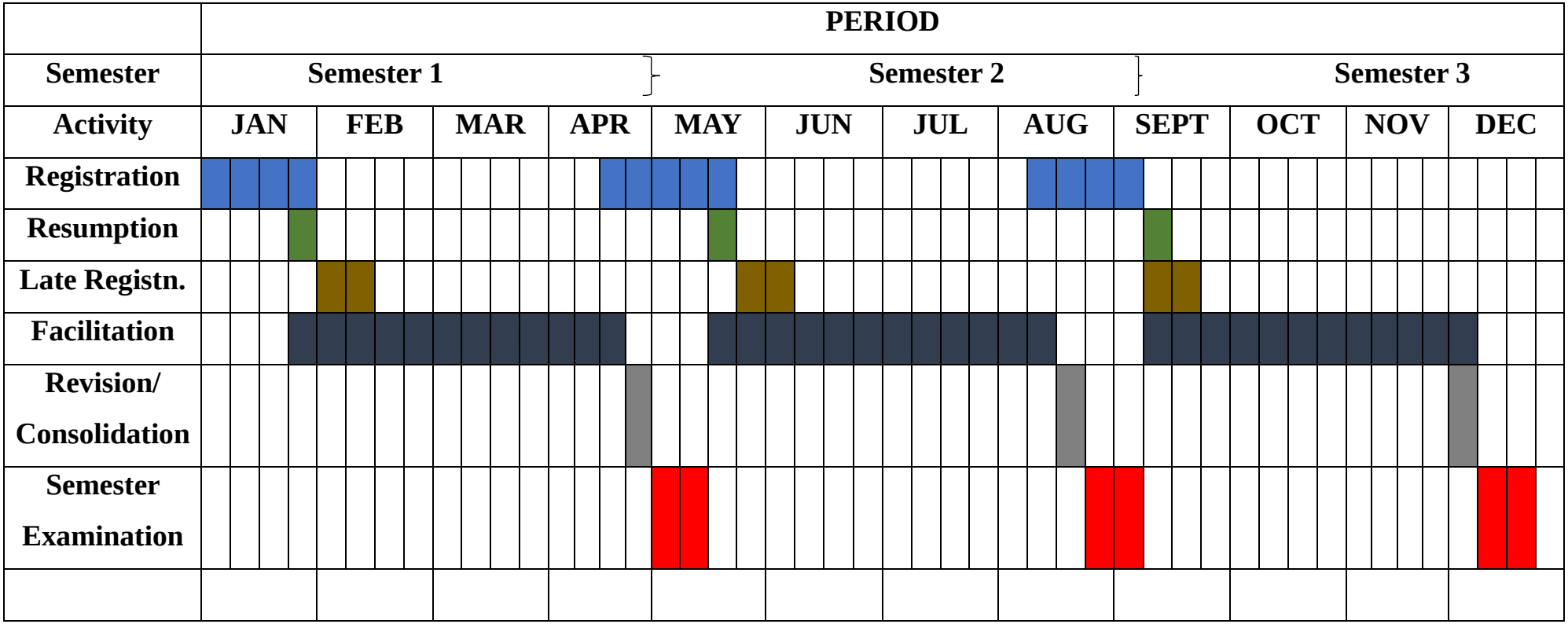
- Unesco's searchable open database is a portal to worldwide courses and research initiatives
- African Virtual University (http://oer.avu.org/) has numerous modules on subjects in English, French, and Portuguese
- https://code.google.com/p/course-builder/ is Google's open source software that is designed to let anyone create online education courses
- Global Voices (http://globalvoicesonline.org/) is an international community of bloggers who report on blogs and citizen media from around the world, including on open source and open educational resources

Individuals (which include OERs)

- Librarian Chick: everything from books to quizzes and videos here, includes directories on open source and open educational resources
- K-12 Tech Tools: OERs, from art to special education
- Web 2.0: Cool Tools for Schools: audio and video tools
- Web 2.0 Guru: animation and various collections of free open source software

- Livebinders: search, create, or organise digital information binders by age, grade, or subject (why re-invent the wheel?)

x. ABU DLC ACADEMIC CALENDAR/PLANNER



N.B: - All Sessions commence in January

- 1 Week break between Semesters and 6 Weeks vocation at end of session.

- Semester 3 is OPTIONAL (Fast-tracking, making up carry-overs & deferments)

xi. COURSE STRUCTURE AND OUTLINE

Course Structure

WEEK	MODULE	STUDY SESSION	ACTIVITY
Week1	**STUDY MODULE 1** Basics of Preservation of Records **Pp.**	**Study Session 1:** What Preservation of Records is all About ***Pp.23***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://youtu.be/8HMp1NKodjI) 5. View referred OER(address/site `https://bit.ly/2Xnq3nH`) 6. View referred Animation (Address/Site https://bit.ly/2K09wBK)
Week 2		**Study Session 2** Knowing the Preservation and close Terminologies of Records ***Pp.35***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2ExQwYo) 5. View referred OER(address/site `https://bit.ly/2EvmWTa`) 6. View referred Animation (Address/Site https://bit.ly/1UeoTmW)
Week3		**Study Session 3** Purpose of Preservation of Records ***Pp.44***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2Efi7Mu) 5. View referred OER(address/site https://bit.ly/2BSV1e6) 6. View referred Animation (Address/Sitehttps://bit.ly/2YLP2Rv)
			1. Read Courseware for the corresponding Study Session.

Week 4		**Study Session 4** Preservation Programmes in Records Centres ***Pp.52***	2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2H0b65k) 5. View referred OER(address/site `https://bit.ly/2T4ORCn`) 6. View referred Animation (Address/Sitehttps://bit.ly/2YGJwj9)
Week 5	**STUDY MODULE 2** Preservation of Records through Fighting Disaster **Pp.**	**Study Session1** Disaster and the Need for Its Management ***Pp.61***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2GKTlHJ) 5. View referred OER(address/site `https://bit.ly/2H3fksU`) 6. View referred Animation (Address/Sitehttps://bit.ly/2TkTGqz)
Week 6		**Study Session2** Planning Activity ***Pp.68***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2Sr9fso) 5. View referred OER(address/site `https://bit.ly/2H3fksU`) 6. View referred Animation (Address/Sitehttps://bit.ly/2T4jE2n)
Week 7		**Study Session3** Prevention Activity ***Pp.75***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2GMQe2b) 5. View referred OER(address/site https://bit.ly/2IEoQVT) 6. View referred Animation (Address/Site https://bit.ly/2JCeQMg)

Week8		**Study Session4** Response and Recovery Activities ***Pp.82***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2XsRsov) 5. View referred OER(address/site `https://bit.ly/2T2qNAb`) 6. View referred Animation (Address/Sitehttps://bit.ly/2HNjiVp)
Week 9	**STUDY MODULE 3 Digital Preservation and Techniques Pp.**	**Study Session1** Strategies of Digital Preservation of Records ***Pp.90***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2tEVix6) 5. View referred OER(address/site`https://bit.ly/2IFFTGU`) 6. View referred Animation (Address/Site https://bit.ly/2W4nOc9)
Week 10		**Study Session2** The Need for Digital Preservation of Records ***Pp.98***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2XsUxoz) 5. View referred OER(address/site `https://bit.ly/2EcGvyl`) 6. View referred Animation (Address/Site https://bit.ly/2EubDKD)

Week 11		**Study Session3** Maintenance of Digital Records ***Pp.103***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2GRL3hB) 5. View referred OER(address/site `https://bit.ly/2IRMuhR`) 6. View referred Animation (Address/Site https://bit.ly/2wfbeaD)
Week 12		**Study Session4** Requirements for a Digital Records Preservation ***Pp.110***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube(address/site https://bit.ly/2tGhsid) 5. View referred OER(address/site`https://bit.ly/2T3Ykdk`) 6. View referred Animation (Address/Sitehttps://bit.ly/30Ewg0e)
Week 13		**REVISION/TUTORIALS (On Campus or Online)& CONSOLIDATION WEEK**	
Week 14 & 15		**SEMESTER EXAMINATION**	

Course Outline

Module 1: Basics of Preservation of Records

Study Session 1: What Preservation of Records is all about?

Study Session 2: Knowing the Preservation and close Terminologies of Records

Study Session 3: Purpose of Preservation of Records.

Study Session 4: Preservation Programmes in Records Centres.

Module 2: Preservation of Records through Fighting Disaster

Study Session 1: Disaster and the Need for Its Management

Study Session 2: Planning Activity

Study Session 3: Prevention Activity

Study Session 4: Response and Recovery Activities

Module 3: Digital Preservation and Techniques

Study Session 1: Strategies of Digital Preservation of Records

Study Session 2: The Need for Digital Preservation of Records

Study Session 3: Maintenance of Digital Records

Study Session 4: Requirements for a Digital Records Preservation

xii. STUDY MODULES

MODULE 1: Basics of Preservation of Records

Contents

Study Session 1: What Preservation of Records is all About?

Study Session 2: Knowing the Preservation and close Terminologies of Records

Study Session 3: Purpose of Preservation of Records

Study Session 4: Preservation Programmes in Records Centres

STUDY SESSION 1

What Preservation of Records is all About

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1 -Preservation of Records and Its Basic Terms

2.2 - Preservation and Forms of Records

2.3 - Preservation of Paper Documents Against Forms of Deteriorations

3.0Study Session Summary and Conclusion

4.0Self-Assessment Questions

5.0Additional Activities (Videos& Out of Class activities)

6.0References/Further Readings

Introduction

You are welcome to our first study session for this module, study session 1will allow you to explore the concept of what preservation of records is all about. You will also explore other aspect concerning preservation of records. This session will an interesting experience. Once again you are welcome.

You can not understand the preservation of records without knowing the basics of the term. This term is needed, because human cultural heritage which is preserved in the records cannot be understood without knowing the history, types, structure etc. of their carriers (records). Therefore, preservation of records is very important to us in the society.

1.0 Study Session Learning Outcomes

After you finished studying this session, you should be able to:

1. Discuss the preservation of records and its basic terms.
2. Explain the preservation and forms of records.
3. Explain the preservation of paper documents against forms of deteriorations.

2.0 Main Content

2.1 Preservation of Records and Its Basic Terms

If we look at this from a different angle preservation of records is of upmost importance in records management. We see that many documents have been lost through time due to torn pages, rust from paper clips, improper storage containers, and exposure to negative elements such as pests, water and heat.

Some suggestions for preserving files include the following:

1. Remove staples, paper clips, binder clips, rubber bands, etc. before storage. Any items other than paper take up unnecessary space and contribute to the deterioration of records.
2. Straighten folded edges and pages.
3. Do not use tape or other mending devices on papers.
4. Avoid using Avery labels or other file labels which may separate from the folder. Handwritten identifiers are more permanent.

5. Make photocopies of pages that are deteriorating through discoloration, folds or tears.

Glossary of Selected Preservation Terms

1. **Conservation –** Physical treatment of items (either individually or en masse) to physically support or chemically stabilize their condition. Treatment actions are based on a prescribed set of ethical and philosophical principles. Treatments result from the collaborative decision-making process that occurs between curatorial staff and preservation personnel. Treatment improves chemical stability.
2. **Damage –** Occurs through neglectful care, use, or handling.
3. **Deterioration –** Occurs as part of the natural aging process. Cannot be halted, but can be retarded through optimal environmental conditions. Selection of chemically stable housing materials can also slow down the rate of deterioration.
4. **Durability –** The ability of a material to withstand physical stresses of use.
5. **Holdings Maintenance –** Basic preservation procedures that are employed to extend the useful life of selected groups of records through proper storage and housing. These procedures help reduce or eliminate the need for intensive, individual attention.
6. **Permanence –** The ability of a material to resist chemical action over time.
7. **Preservation –** (Umbrella term) denotes the wide variety of preventive and remedial activities performed to maintain one's holdings. Holdings may be preserved in their original or other usable format. Activities provide protection for items currently housed in or added to our holdings. (e.g. environmental monitoring, reformatting, re-housing, conservation, etc.).

8. **Preservation Program –** Formally established, directed and coordinated set of activities associated with maintaining materials for use based on institutional priorities.
9. **Reformat –** Action taken to preserve information. Most common reformatting processes for the printed word are currently microfilming and electrostatic photocopying.
10. **Repair –** Action taken in good faith to improve the condition of the record. In the long run, many well-intentioned, but un-informed "repairs" do more harm than good. Occurs as a subset of conservation when appropriate techniques and materials are used.
11. **Restoration –** Action taken to create a known or assumed state. Primary focus is placed on appearance rather than structural stability. Often has a derogatory connotation for archives materials.

In-text Question 1
Itemize some of the Selected Preservation Terms

Answer
1. *Restoration*
2. *Repair*
3. *Reformat*
4. *Preservation Program*
5. *Permanence*
6. *Preservation*
7. *Conservation*

2.2 Preservation and Forms of Records

This means protecting, and thus preserving, the record through appropriate storage facilities and conditions, ensuring that the record is handled appropriately by staff, and protecting the records if a disaster should happen.

It is important for you to assist public offices in understanding and implementing this obligation for physical hard-copy records, the revised Standard on the physical storage of State records sets out the minimum

compliance requirements for the storage of those State records which have a physical format, including:

1. paper files and documents
2. volumes and registers
3. maps, plans, charts and drawings
4. photographic media including photographic prints and negatives, film, microforms, and x-rays
5. magnetic media such as digital tape, video and audio cassettes, and
6. Optical media such as CDs and DVDs.

You should know that this standard does not cover the storage of digital records in online systems or network servers.

The storage and preservation of physical records is a key element in the records management program of all public offices.

Fig 1.1.1: Preservation and Forms of Records

You observe that organisations are encouraged to develop a comprehensive storage plan covering all types of records and their storage locations and ensure that appropriate arrangements for storage have been made to ensure the security, maintenance and accessibility of the organisation's records. We also see that all public sector organisations should ensure that the physical storage of records is effective in protecting records from deterioration and that the storage of records is cost efficient.

In-text Question 2
What are some examples of preservation terms?

Answer
1. Conservation
2. Damage
3. Deterioration and
4. Repair etc.

2.3 Preservation of Paper Documents Against Forms of Deteriorations

Paper-based materials can deteriorate chemically and physically. Some deteriorate because of their inherent properties, others as a result of poor storage or display conditions. We observe that damage can be caused by poor handling practices. The media on paper-based objects, such as ink, watercolour or pencil can also degrade. While we cannot stop deterioration, we can influence the rate at which it happens.

Deterioration

Let us discuss different forms of deterioration

Chemical deterioration includes:

1. Fading and discolouration of papers and the media on the paper caused by exposure to light or ultraviolet (UV) radiation. Deterioration will occur more quickly when exposed to higher light intensity.
2. Deterioration caused by mould attack that can develop in humid conditions; mould has the ability to consume and stain the materials on which it grows.
3. Paper brittleness and staining caused by acid either in the paper or the material with which the paper is in contact.
4. Deterioration caused by interaction between the paper and the printing medium used. Iron gall ink is acidic by nature and over time, can burn through the paper on which it is written.

Physical deterioration includes:

1. handling and use damage such as, tears, dents, punctures, abrasions, and paper losses
2. folds and creases causing paper to split
3. pest attack, caused by some insects and rodents for which paper and glues are a food source

4. Extreme temperature and relative humidity fluctuations that cause buckling and distortion.

Correct handling of flat paper items will aid their long-term preservation. The following procedures should be followed:

1. Always handle with care.
2. Never allow food or drink near items
3. Have clean hands, even when wearing protective cotton or powder-free surgical gloves. If an object is particularly fragile close-fitting surgical gloves may be more appropriate.
4. Use a rigid support, such as piece of cardboard when carrying papers. This is important where papers are oversized and fragile or need to be carried over long distances.
5. A flat-bed trolley is useful for moving oversized material.
6. Interleave artworks and documents with archival tissue to protect them from abrasion and ink or adhesive transfer from other items.
7. When working with artworks, use pencil not ink. Pencil marks are readily removed, whereas inks can be difficult or impossible to remove.

Preservation and Storage

The following are the factors to be considered in the course of preservation and storage of records:

Environment

If specialised storage is available then temperatures between 18 to 20 °C and at relative humidity between 45 to 50 per cent are desirable. Office air-conditioning generally operates between 21 and 23 °C in temperate climates. If these conditions are stable then the natural deterioration of the objects can be

moderated. Wide fluctuations in conditions can physically stress records and +accelerate their deterioration.

Note that air-conditioning systems usually only operate during office hours, leading to a wider fluctuation of temperature and relative humidity at other times. At sustained relative humidity over about 65 per cent, mold growth can occur. It is recommended that you aim for a storage area with stable conditions, where the temperature is around 20 °C and the relative humidity remains below 60 per cent.

Corrugated board storage boxes can provide a convenient micro environment by buffering documents from the full effects of fluctuating humidity and temperature.

This should not be new to you, but materials degrade more quickly when exposed to light, especially ultraviolet (UV). Surprisingly, we found out that fluorescent tubes often emit a relatively high level of UV. Your lighting should be turned on only when it is needed. Ideally, storage areas should have no windows. Where windows are present, they should be covered with opaque curtains or blinds.

You may have experience it at a point in time, insects and rodents will cause damage through what they eat or leave behind. You should reduce the chance that they will be attracted to record storage areas by following these recommendations:

1. Do not eat in storage areas.
2. Keep surfaces (floors, tops of shelves) clean.
3. Empty rubbish bins regularly.
4. Check storage areas regularly for outbreaks of insects or rodents. Should an infestation occur, baiting or fumigation may be necessary. Further advice is available on integrated pest management.

Shelving

You should use properly cured powder-coat painted metal shelves for paper records and plan cabinets for flat storage of maps, plans and large artworks. Your prints, drawings, posters and watercolours should be stored flat, while paper files are usually stored vertically. Clearly label storage boxes and folders to minimize unnecessary handling. The storage area and facilities should be clean to discourage pests. Unsealed wooden and particle board/MDF shelving can release harmful vapours and should be replaced where possible.

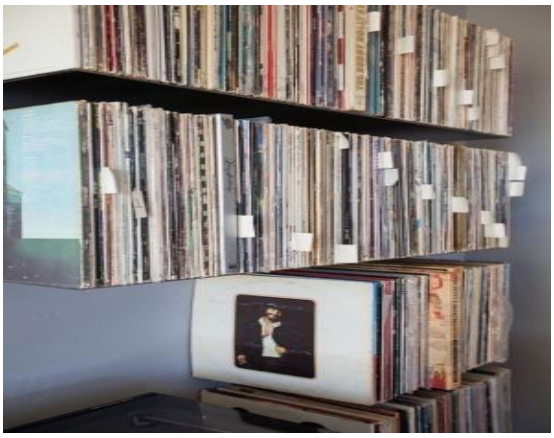


Fig 1.1.2: Shelving

Packaging

We found out that, simple archival enclosures such as boxes, folders, wallets and paper cards protect paper based items against mechanical damage, light and dust. They also provide micro-environments that reduce temperature and relative humidity fluctuations.

Choose a high-quality paper product for long-term storage. The Archives has established a list of local suppliers of archival materials. Some plastics are also suitable for paper record storage. The plastic should be free of plasticisers, surface coatings and other harmful chemicals.

The most suitable plastic film is a form of polyester known as PET, which sells under the proprietary names Mylar or Melinex. Food grade polypropylene and polyethylene are acceptable lower cost alternatives.

We see that encapsulation is a method of storage that uses a close fitting stiff polyester sleeve to take the majority of handling pressure and abuse. It allows an item to be viewed in its housing without direct contact. Hot lamination where the plastic sleeve is adhered to the original documents and artworks is not

recommended. We found out that this treatment is irreversible and will lead to the accelerated deterioration of the paper inside.

If you require further conservation advice on the storage and display of paper documents and artworks contact the Agency Service Centre.

In-text Question 3
What is preservation?

Answer
Answer: Preservation means protecting the record through appropriate storage facilities and conditions, ensuring that the record is handled appropriately by staff, and protecting the records if a disaster should happen.

3.0 Study Session Summary and Conclusion

This session presented the preservation of records and its basic terms, preservation and forms of records as well as preservation of paper documents against forms of deteriorations. It is concluded that understanding the preservation of records cannot be achieved without knowing its basic terms.

4.0 Self-Assessment Questions

1. What is preservation of records?
2. What are the forms of records?
3. How is the preservation of paper documents being done against forms of deteriorations?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://youtu.be/8HMp1NKodjI. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2K09wBK and critique it in the discussion forum

6.0 References/Further Readings

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STUDY SESSION 2
Knowing the Preservation and Close Terminologies of Records

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.2 - Common Terminologies Used in Place of Preservation of Records

2.2 - Principles of Arrangement of Records and Terminologies

2.3 - General Principles of Records Preservation and Terminologies

3.0 Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos & Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome to study session 2, in the previous session we took a deep dive into preservation of records, this session will look at the concept of preservation and close terminologies of records. Once again welcome.

You will agree with me that, preservation of records is understood through basic terminologies. The terminologies are sometimes used in place of the word preservation itself. They are conservation and restoration. We see that these terms are some of the basic concepts of preservation of records.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the common terminologies used in place of preservation of records
2. Explain the principles of arrangement of records and terminologies.
3. Discuss the general principles of records preservation and terminologies

2.0 Main Content

2.1 Common Terminologies Used in Place of Preservation of Records

If we are to look back at history, Since the 1980s, we found out that the library and archival communities have used "**preservation**" as an umbrella term for activities that reduce or prevent damage to extend the life expectancy of collections, while "conservation" refers more specifically to the physical treatment of individual damaged items. "**Preventive conservation**" is another term that you can use to describe broad collections care activities that support the longevity of artefacts and records, such as environmental monitoring. You use the term "restoration" mostly in the context of museum objects or motion picture films. It generally refers to the process of returning an object to its original state, or what is thought to have been its original state.

Fig 1.2.1.: Common Terminologies Used in Place of Preservation of Records

The American Institute for Conservation of Historic and Artistic Works (AIC) Definitions of Conservation defines these three terms as follows:

1. **Preservation**—The protection of cultural property through activities that minimize chemical and physical deterioration and damage and that prevent loss of informational content. The primary goal of preservation is to prolong the existence of cultural property.
2. **Conservation**—The profession devoted to the preservation of cultural property for the future. Conservation activities include examination, documentation, treatment, and preventive care, supported by research and education.

3. **Restoration**—Treatment procedures intended to return cultural property to a known or assumed state, often through the addition of non-original material.

In the widest sense, preservation encompasses

1. activities that prevent damage to paper-based and audio-visual collections, such as proper housing, environmental control, and disaster planning; and
2. Activities such as treatment, replacement, or reformatting that address existing damage.

Using the term **preservation**, we mean keeping a balance between collection-level activities such as environmental control, which can be difficult and/or costly to manage but provide the greatest long-term benefit for the most materials, and item-level activities such as conservation treatment, which are often more easily understood and managed but can have limited effect, especially if the items are returned to a damaging environment.

In-text Question 1
What is Preservation?

Answer
Preservation is the protection of cultural property through activities that minimize chemical and physical deterioration and damage and that prevent loss of informational content.

2.2 Principles of Arrangement of Records and Terminologies

The following are the principles of arrangement of records and terminologies:

Arrangement of Records Groups

In the National Archives an initial determination on the arrangement to be given records is made at the time they are allocated to records branches. The

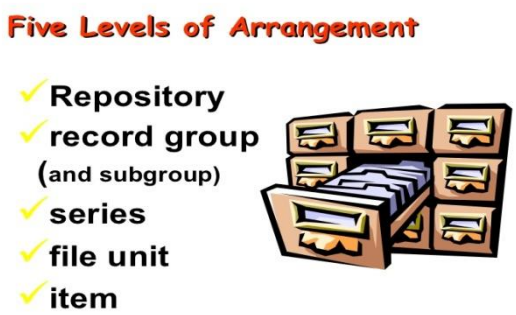

Fig 1.2.2: Arrangement of Records Groups

allocation of records is made on the basis either of their relation to some broad subject-matter field (such as defence, industry, or natural resources) or of their technical character (such as cartographic or audio-visual). Subject-matter relationships, however, are defined at this stage mainly in terms of the functions of the agencies that created the records. Thus records created by the Department of Agriculture and by independent agencies concerned with agricultural activities are allocated to the Natural Resources Records Branch and within this branch to the Agriculture Records Section.

Arrangement of Sub-groups

Once a plan has been chosen for the arrangement of record groups showing their relations to each other, the next step is for you to provide the arrangement within each record group of its subgroups. Arrangement according to a scheme is more important at this level than at the level of the record group.

As in the case of the arrangement of record groups, the subgroups should be placed, insofar as possible, in logical relation to each other -- according to hierarchy, chronology, function, geographical location, or subject.

Arrangement by Series

Within the subgroups, series should be arranged according to some logical pattern that reflects the interrelationships among series and where appropriate the relationships of series to organisation, functions, chronological periods, places, or subjects. The ultimate physical arrangement of series on shelves should be anticipated as far as possible when records are initially stored, and this arrangement should be carried out so that when a well-organised inventory has been prepared it will correspond to the physical arrangement of the records.

Arrangement of File Units

The final, and most detailed, step in arranging records is concerned with single documents, folders, dossiers, volumes, or other file units. File units, it was noted in Staff Information Paper No. 17, usually consist of records kept together because they relate to the same subject or transaction, or because they have the same form. You observe that these units, which vary in size and character, are usually placed in a sequential arrangement that is determined by the type of filing system employed.

In a subject system whether it is arranged on an alphabetical, a subject-numeric, a classified, or some other basis records will ordinarily be filed together under subject captions, each of which may cover a folder or several folders, which, in turn, may contain a number of separate documents. In a case file system whether it is arranged alphabetically, numerically, or in some other way records will be assembled in case folders or dossiers.

Where records are kept together because of similarity of form, the units of form will often be considered as the file units. This is the case with respect to bound volumes.

In-text Question 2
What are the principles of arrangement of records?

Answer
1. Arrangement of records groups
2. Arrangement of sub-groups
3. Arrangement by series and
4. Arrangement of file units.

2.3 General Principles of Records Preservation and Terminologies

The framework, which is also known as simply the Principles, has eight precepts for creating information governance best practices:

1. **Principle of Accountability -** An organisation shall assign a senior executive who will oversee a recordkeeping program and delegate

program responsibility to appropriate individuals, adopt policies and procedures to guide personnel and ensure program audit ability.

2. **Principle of Transparency-** The processes and activities of an organization's recordkeeping program shall be documented in an understandable manner and be available to all personnel and appropriate interested parties.
3. **Principle of Integrity -** A recordkeeping program shall be constructed so the records and information generated or managed by or for the organization have a reasonable and suitable guarantee of authenticity and reliability.
4. **Principle of Protection -** A recordkeeping program shall be constructed to ensure a reasonable level of protection to records and information that are private, confidential, privileged, secret, or essential to business continuity.
5. **Principle of Compliance -** The recordkeeping program shall be constructed to comply with applicable laws and other binding authorities, as well as the organization's policies.
6. **Principle of Availability -** An organisation shall maintain records in a manner that ensures timely, efficient, and accurate retrieval of needed information.
7. **Principle of Retention -** An organisation shall maintain its records and information for an appropriate time, taking into account legal, regulatory, fiscal, operational and historical requirements.
8. **Principle of Disposition -** An organisation shall provide secure and appropriate disposition for records that are no longer required to be maintained by laws and organisational policies.

In-text Question 3
What are the general principles of records preservation?

Answer

Principle of accountability, principle of transparency, principle of integrity, principle of protection, principle of compliance, principle of availability, principle of retention, principle of disposition.

3.0 Study Session Summary and Conclusion

This session presented the common terminologies used in place of preservation of records, the principles of arrangement of records and terminologies and the general principles of records preservation and terminologies.

4.0Self-Assessment Questions

1. What are the common terminologies used in place of preservation of records?
2. What are the principles of arrangement of records?
3. What are the general principles of records preservation?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2ExQwYo. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/1UeoTmW and critique it in the discussion forum

6.0 References/Further Readings

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STUDY SESSION 3

Purpose of Preservation of Records

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1 - Reasons for Preservation of Records

2.2 - Preservation of Records as a Way of Achieving General Records Management

2.3 - Achieving the Purpose of Preserving Records through Protective Measures

3.0Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos & Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome to study session 3, the two previous session we look at the concept of preservation of records, this session will explore the purpose of preservation of records.

Preservation of records plays major role in your society. You see that the role it plays cuts across all the disciplines of human endeavours. For instance, in agriculture, health care, education, security, transportation to mention but few, preservation of records helps in accessing the information needed for their management by the stakeholders responsible for that.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the reasons for preservation of records.

2. Explain that preservation of records is a way of achieving general records management.
3. Explain that the purpose of preserving the records is achieved through protective measures

2.0 Main Content

2.1 Reasons for Preservation of Records

Compendium of Information: They are the materials that reflect social and economic and political life e.g. Books, journals etc. which are resources of information therefore they should be preserved for this reason.

Raw Materials of History: Records and archival materials provide background information about an event in history such material containing that information should be preserved for the future users who might need that information for example the national anthem of the country.

Wide Range of Users: Preservation practices like photo copying of the materials and digitization will help to make many copies of the document and different formats of the records, these make it possible for many people to access the information thus making preservation necessary.

Capturing of Information: For example everyone from a child to an old man turns to information after 100 years there for the information must be preserved for the next generation like the land title certificate, bibliography of the people of the clan.

For Future Use and Reference: For example the birth certificates should be preserved for use in future when the owner grows up like for registration and during job interviews therefore hard copies of the old as well as new copies are prove to decay so to provide continuous and wider access, preservation is a must.

Rare Materials: Manuscripts and other material are of immense values from cultural and historical point of view like the kings letters, presidential mails

therefore they need to be preserved. Preservation seeks to reduce risk of damage and to slow down the rate of deterioration of records through processes such as controlling light that speeds up the oxidation of paper, causing materials to deteriorate faster.

Preservation of information and **archival management** is carried out to prevent information materials from fading and discoloration of papers and media materials these would lead to loss of information they contain so there is need to preserve this information. Preservation of information and archival management aims at increasing the life expectancy of information materials through fumigation, lamination, photocopying of these materials which increases the time they spend with the information being lost.

Fig 1.3.1: Reasons for Preservation of Records

Preservation is useful in long-term protection of information material against decay, inappropriate temperature, inserts since these process take long when there effective on the preservation of the record. Preservation Increases the depth of the collection and improves access to various subjects for example, the use of indexes and catalogue makes it easy to access the information you want.

Preserving library collections promotes democracy to people in order to access information at no cost these will make people to enjoy their right to information for instance classification process of the collection. To safe guard the document information from theft of the material and to reduce the threat of fire and water damage among others.

To avoid inappropriate mending or repair work which may lead to the destruction of the document and the information will be used forever so preservation should be carried out.

To safe the archival material from being attacked by biological agents such as mold, insects or rodents these would destroy their material which is very difficult to find again now days since the technology has changed so there is need to safe guard these materials by preservation.

In-text Question 1

List the reasons for preservation of records

Answer

1. *Compendium of Information*
2. *Raw Materials of History*
3. *Wide Range of Users*
4. *Capturing of Information*
5. *for Future Use and Reference*
6. *Rare Materials*

2.2 Preservation of Records as a Way of Achieving General Records Management

You observe that records management helps organisations classify, store, secure and destruct or preserve records. There are a lot of reasons for your organisation to develop and implement a records management program, and here are five of them:

Control over the Creation and Growth of Records

An effective records management program does two major things for you. It limits how many records and copies of records are created that are not essential to the business. And, it serves as a system for destroying useless or retiring inactive records.

Preserve Corporate Memory

Your business records contain your organisations memory, an irreplaceable asset that is often overlooked. Your records are created each day that become background data for future business decisions and planning. It is important that your records management they are preserved securely.

Improve Efficiency and Productivity

Your well organised records manage make it easier and quicker for employees to access them when they are needed and files them correctly when they are not. This not only improves efficiency and productivity but it also saves time and space.

Ensure Compliance

We have previously discussed that the US is one of the most heavily regulated countries in the world. Our laws can create major compliance problems for businesses since they can be difficult to interpret and apply. One way an organisation can be reasonably sure that it is in full compliance is by operating a good records management program which takes responsibility for regulatory compliance.

Protect Important Information

We have discussed that every organisation needs to protect its records from disaster because every organisation is vulnerable to loss. Your successful records management program preserves the **integrity** and **confidentiality** of all important records.

In-text Question 2
What are the reasons for preservation of records?

Answer
1. Compendium of information
2. Raw materials of history

3. Wide range of users,
4. Capturing of information, for future use and reference and finally for rare materials

2.3 Achieving the Purpose of Preserving Records through Protective Measures

We have come to understand that it is very important that agencies should develop appropriate protective measures for their records and copies of their vital records to respond to actual or potential emergencies or disasters identified in contingency planning. This is the **records management** aspect of emergency management. Vital records are emphasized because they tend to have the greatest value in case of emergency or they require extra protection because they document legal or financial rights.

The type and level of value determine the amount of protection agencies should provide. Special protective measures for vital records may include using fire-rated filing equipment for storage; constructing onsite vaults; transferring records to offsite storage; duplicating the records at the time of their creation, such as computer "backup" tapes, using existing duplicates as vital record copies; or microfilming vital records.

You should employ more additional protective measures for Federal records maintained on a medium other than paper. You should keep in mind that, these "special records" require specific environmental conditions and careful handling throughout their life cycle to ensure their preservation. Your agencies must maintain temperature and humidity controls for special records such as photographs and negatives, microforms, audio, and video tapes and disks, and electronic tapes and disks.

You will agree with me that, when emergencies or disasters occur, however, even the best of protective measures may not prevent damage to records. Consequently, agencies need to develop records recovery plans for timely and

economical response to records disasters in order to salvage or replace damaged records and the information that they contain.

In-text Question 3
What are the common preservation practices that allow wide range of users access to records?

Answer
1. Photocopying and
2. Digitisation.

3.0 Study Session Summary and Conclusion

This session presented the reasons for preservation of records, preservation of records as a way of achieving general records management as well as the protective measures that help in achieving the purpose of preserving records.

4.0 Self-Assessment Questions

1. What are the reasons for preservation of records?
2. How the preservation of records cans leads to the achievement of general records management?
3. What are the protective measures that help in achieving the purpose of preserving records?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2Efi7Mu. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2YLP2Rv and critique it in the discussion forum

6.0 References/Further Readings

Dennis, K. (2014). 5 Reasons for Records Management. Retrieved from http://community.aiim.org/blogs/dennis kempner/2014/02/20/5-reasons-for-records-management. Accessed on 11/1/2019.

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STUDY SESSION 4
Preservation Programmes in Records Centres

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1 - Building a Preservation Programme

2.2 - Basic Elements of Preservation Programme

3.0Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos & Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome study session 4, we are still on the topic of preservation of records, just as we have been discussing in the three previous session. In this session we will explore the concepts of preservation programmes in records centres.

We should have it in mind that preservation programmes in records are important. They help in achieving the aims and objectives of such centres. Since preservation of records is a continuous process, is needs to be carried out in a systematic way. The systematic way of preservation of records is done through programmes and plans together with the requirements (security, environmental control, conservation treatment etc.) needed.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the term building of a preservation programme.
2. Explain the basic elements of a preservation programme

2.0 Main Content

2.1 Building a Preservation Programme

This section presents the process of building a preservation programme as follows:

The Planning Process:

The basic components of the preservation planning process are:

1. identify and assess preservation needs (through some type of survey)
2. set priorities among preservation needs according to the institution's mission
3. evaluate existing preservation activities
4. decide what preservation functions are needed within the institution and how they will be organized
5. allocate resources for preservation
6. write a plan and determine how the plan will be implemented

Identifying and assessing needs is an information-gathering process: what collections does the institution hold and what is their condition? How are collections housed? What are the environmental conditions within the building(s)? Is there a disaster plan? Are collections protected from fire and theft? Are repair, binding, and/or conservation treatment undertaken regularly? How much funding and staff time is allocated for preservation?

Once you have discerned your preservation needs, you must consider potential solutions to the problems you have identified. What methods are most appropriate for your particular institution and situation? What existing activities already have a preservation component, and how should existing preservation activities change to meet the needs you have identified?

As part of this process, you will need to assess your institutional mission and goals and the personnel and monetary resources currently available, as well as potential future resources.

Once all the relevant issues have been considered, you should set preservation priorities and pull together a preservation strategy that includes an action plan and timetable. A systematic preservation plan will allow you to respond effectively to preservation needs and take advantage of preservation opportunities that might otherwise have been missed.

Preservation Strategies

The following are the preservation strategies:

Developing a Preservation Strategy

While the basic elements of a preservation program are fairly straightforward, the emphasis placed on each of these elements will vary according to the institution and situation. The relative importance of each component for your institution will depend on the type, value, and condition of your collections; the importance of specific collections to your institution's mission; the type of access you need to provide (open stacks, closed stacks, offsite storage); and the amount of use the collections receive.

Fig 1.4.1: *Developing a Preservation Strategy*

You will need to develop an appropriate preservation strategy for your institution. In general, preservation program elements that benefit all collections (such as environmental control, disaster planning, and proper storage and handling) should be thought of as the basic "umbrella" components of a program. Other strategies, such as reformatting or conservation treatment methods, may be additional considerations.

Strategies for Different Types of Institutions

Preservation needs and responses are specific to a particular institution and each collection within that institution. Some repositories may see their primary responsibility as preserving unique and/or comprehensive collections over the long term, while others may place more emphasis on serving academic teaching and research needs, or services to the general public.

In collecting archival records and artefacts, cultural heritage organisations assume an obligation to preserve those collections over the long term. These institutions will place a strong emphasis on strategies to stabilise the collections as a whole (e.g., holdings maintenance/collections care, environmental control, security, emergency preparedness), with a secondary focus on preservation reformatting (usually digitisation) and/or conservation treatment for the specific collections that require it.

The primary mission of small- to mid-sized public or academic libraries, on the other hand, is to provide you with access to collections, for as long as the materials are needed. These libraries tend to have a high level of use and collect basically rather than comprehensively. In these libraries, reference collections and materials related to current courses receive the most use. These institutions will need to focus on umbrella issues such as storage and handling, environment, security, and disaster planning, but in addition they will undertake

library binding, basic book repair, withdrawal of brittle titles that are no longer needed, and replacement of brittle titles with reprints or digital versions. Preservation reformatting and conservation treatment will generally not be a priority, unless the institution also holds historical or other unique collections.

Like smaller academic and public libraries, large research libraries strive to provide users access to collections for as long as the materials are needed. In addition, they assume responsibility for building and maintaining research collections that are meant to survive over the long term. For these institutions, an even wider range of preservation options will be needed, including umbrella strategies such as environmental control, storage/handling, and disaster planning, as well as preservation reformatting, replacement, library binding, book repair, and conservation treatment.

In-text Question 1
What is the first step of building a preservation programme?

Answer
1. Planning process

2.2 Basic Elements of a Preservation Programme

Environmental Control—providing a moderate and stable temperature and humidity level as well as controlling exposure to light and pollutants. This should be a priority for all institutions, although control will usually be less tight for general circulating collections than for rare books, special collections, or archival materials.

Disaster Planning—preventing and responding to damage from water, fire, or other emergency situations. Again, you should make it a high priority in all institutions. The reasons are obvious, for collections of enduring value, but even collections that are not meant to be retained over the long term represent a capital investment for an institution and as such must be protected from loss.

Security—protecting collections from theft and/or vandalism. This type of protection is needed for both special and general collections, since loss and

vandalism of general collections results in unnecessary replacement and expense.

Storage and Handling—using non-damaging storage enclosures and proper storage furniture; cleaning storage areas; using care when handling, exhibiting, or reformatting collections and educating staff and users in proper handling techniques. Again, you should make it a priority for all types of collections.

Reformatting—reproducing deteriorating collections onto stable media to preserve the informational content or in cases where the originals are fragile or valuable and handling is restricted. This category includes microfilming, production of preservation facsimiles, and duplication of audio-visual collections. These strategies are most appropriate for collections whose intellectual content needs to be preserved over the long term and/or where security copies are needed for unique items.

Preservation microfilming is still an effective strategy for unique paper-based collections, but a low priority for institutions with general collections that are duplicated elsewhere.

Library Binding—rebinding of damaged volumes to provide sturdy use copies. This strategy is used by libraries with general collections in heavy use. It should not be used on any items that have art factual value.

Conservation Treatment—treating individual objects using the services of a trained conservator. This may be appropriate for a wide range of institutions, provided they hold unique materials that are of sufficient value to justify treatment.

In-House Repair—repairing objects that do not have artifactual value using a trained collections conservator or trained in-house staff. In-house book repair is used by public and academic libraries to keep non-unique books in good condition for use, and some institutions use basic paper repair techniques (e.g., mending, encapsulation) for historical materials. For special collections

libraries, archives, and historical societies, general preventive activities such as rehousing should be given a higher priority than in-house repair.

Digital Reformatting and Preservation—using digital imaging to provide access copies of deteriorated original collections; creating digital objects that will act as preservation copies of original items; and/or preserving objects that are "born-digital." Digital projects may be appropriate across a wide range of institutions; the key in undertaking such a project is for the institution to have a good understanding of the requirements and limitations of digital imaging.

In-text Question 2
What are the basic elements of a preservation programme?

Answer
Environmental control, disaster planning, security, storage and handling, reformatting, library binding, conservation treatment, in- house repair and digital reformatting and preservation

3.0 Study Session Summary and Conclusion

This session presented the term building of a preservation programme as well as the basic elements of a preservation programme.

4.0Self-Assessment Questions

1. What is the meaning of the term building of a preservation programme?
2. What are the basic elements of a preservation programme?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2H0b65k. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2YGJwj9 and critique it in the discussion forum

6.0 References/Further Readings

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STUDY MODULE 2

Preservation of Records through Fighting Disaster

Contents

Study Session 1: Disaster and the Need for Its Management

Study Session 2: Planning Activity

Study Session 3: Prevention Activity

Study Session 4: Response and Recovery Activities

STUDY SESSION 1

Disaster and the Need for Its Management

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1 - Disaster in the Context of Preservation of Records

2.2 - Risk Assessment for Records Preservation

3.0Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos &Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome to study session 1, in this session our focus of interest is to discuss and explore the concept of disaster and its management. Once again you are welcome.

This should not be new to you. As you may know, disaster is an enemy to records. It has been destroying records for a very long period of time. As such, fighting it is necessary in ensuring preservation of records irrespective of the

type, environment, and magnitude among others.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the term disaster in the context of preservation of records.
2. Explain risk assessment for records preservation.

2.0 Main Content

2.1 Disaster in the Context of Preservation of Records

Let us define what disaster is. You can define disaster as a "sudden or great misfortune" or simply "any unfortunate event." More precisely, a **disaster** is "an event whose timing is unexpected and whose consequences are seriously destructive." These definitions identify an event that includes three elements:

Fig 2.1.1: Disaster in the Context of Preservation of Records

1. Suddenness
2. Unexpectedness
3. Significant destruction and/or adverse consequences

However, a fourth element, lack of foresight or planning, is sometimes added. Disasters occur with unnerving frequency. Their adverse consequences increase for those who do not prepare for predictable contingencies. A disaster prevention and recovery plan can help protect all of the University's assets including people, jobs, records, vital records, and facilities.

Disasters are not restricted to records and information resources. The death of an essential employee, a poisoning, an explosion, a fire, or a chemical spill are disasters that adversely affect the University. The University, as a whole, must protect all of its assets. Your plan must be tailored to meet the needs of your department, facility, and types of information.

The plan can easily get out of hand by adding unnecessary costs which will make the plan appear unfeasible to management. Your plan must include a clear definition of your department's records (records refer to all recorded information, regardless of the media or characteristics) to be protected in the disaster plan. Elements to avoid in your plan should include the reconstructing or salvaging of reference material, convenience copies, and non-essential files.

University records vary greatly in value. Whether stored electronically or on paper, the plan must identify historical and vital records records that document your department's origin, growth, development, operations, and civic contributions as well as records that are essential to the resumption and continuation of business after a disaster. A current list of vital records is necessary to determine the extent of any **information disaster**. Procedures for protecting and reconstructing vital records stored on magnetic media, optical disk or microforms differ from those protecting and salvaging information contained on paper. Disaster plans must include and provide for all media on which records are created and stored.

In-text Question 1
What is disaster?

Answer
A disaster is defined as a "sudden or great misfortune" or simply "any unfortunate event." More precisely, a disaster is "an event whose timing is unexpected and whose consequences are seriously destructive."

2.2 Risk Assessment for Records Preservation

Completing a Risk Assessment involves you determining the probability of a particular disaster occurring in your office and the effects that the disaster may

have on the operations of your office or your records. A Risk Assessment also helps you determine which protection method is best for your records. Although this is largely an exercise in probability, since we never know what will happen, it will narrow the scope of protection methods and allow for some early disaster preparedness.

There are 3 basic steps to completing a risk assessment:

1. Identify the risks your office may encounter
2. Determine what level of impact the risk will have
3. Calculate the probability of that risk happening

First you will need to identify the 5-6 greatest risks to the records of your particular office. Not all offices are likely to face the same risks, although fire and water damage are the most common. For example, an office that deals in research may have the added risk of sabotage, whereas an office dealing strictly with technology and computers would have a far greater risk of losing information in a power outage or hard drive crash.

After you have identified the top risk factors that may affect your office, list the individuals who can help you with recovery from that risk. This could include Capital Projects, Physical Plant, OIM, or UW Technology staff. If no one in your office has the necessary skills, consider contracting the services of a vendor.

There are three categories of disasters:

Natural Disasters

1. Flooding
2. Fire
3. Earthquake
4. Wind damage

5. Snow/ice storm
6. Volcanic eruption
7. Electromagnetic interference
8. Vermin/insects
9. Hurricane

Technical Disasters

1. Power failure
2. HVAC failure
3. Malfunction or failure of CPU
4. Failure of system or application software
5. Telecommunications failure
6. Gas leaks
7. Communications failure

Human Disasters

1. Data entry error
2. Improper handling of sensitive data
3. Unauthorised access
4. Malicious damage or destruction of data
5. Robbery/theft/burglary
6. Bomb Threats
7. Epidemic
8. Explosion
9. Strikes/picketing
10. Civil disorder
11. Loss of physical access to resources
12. Chemical spill
13. Vandalism/Sabotage

14. Hazardous material
15. War

In-text Question 2
What are the categories of disasters?

Answer
1. Natural disasters
2. Technical disasters and
3. Human disasters

3.0 Study Session Summary and Conclusion

This session presented the term disaster in the context of preservation of records as well as risk assessment for records preservation. They are presented in order of their arrangement.

4.0 Self-Assessment Questions

1. What is the meaning of the term disaster in the context of preservation of records?
2. What is risk assessment for records preservation?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2GKTlHJ. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2TkTGqz and critique it in the discussion forum

6.0 References/Further Readings

Minnesota Historical Society.(2013). Disaster Preparedness. Retrieved from http://www.mnhs.org/preserve/records/disaster.php. Accessed on 24/1/2019.

The Curators of the University of Missouri. (2011). What is a Disaster? Retrieved from

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STUDY SESSION 2
Planning Activity

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1 - The Basics for Disaster Plan

2.2 - The Need for the Development of Disaster Plan

2.3 - Disaster Planning Requirements in Records Centres

3.0Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos & Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome to study session 2, this session discussion will be on the concept of planning activity. You will also learn the need for planning as in regard to disaster.

Planning is your first stage of preservation of records. The goal of preservation of records cannot be achieved without planning from the beginning. Planning helps you in identifying the records to be preserved, the reason for their preservation, the time for the preservation, the place of preservation, the procedure of preservation, the facilities, equipment etc. needed for preserving the records.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the basics for disaster plan.
2. Explain the need for the development of disaster plan.
3. Explain the disaster planning requirements in records centres.

1.0 Main Content

2.1 The Basics for Disaster Plan

The first element in formulating a records disaster prevention and recovery plan is your clear understanding of disasters themselves - their types, causes, and effects. A plan tailored to your needs can only be accomplished by a team or committee, or someone hired from outside to come in and study your operation. We, who gather information concerning the contents of any particular disaster plan, can only speculate as to what will or will not actually be needed in your department's plan. Completion of the study is easier when members of your department are involved in all stages of writing the plan. A plan based on anything other than your needs will result in a weak, costly plan that does not meet the criteria of effectiveness and efficiency.

2.2 The Need for the Development of Disaster Plan

Disasters come in many flavours. Unfortunately, we do not get to choose our favourite flavours, nor do we get to select the time when we will be ready to sample their varieties. By their very nature, disasters are sudden, unexpected, and unwanted. Some, such as those resulting from faulty wiring or leaky plumbing, can be avoided;

Fig 2.2.1: The Need for the Development of Disaster Plan

others, such as floods, tornados, and winter storms, cannot. One aim of this guide is to assist individuals in preventing disasters from becoming catastrophes events that result in great destruction or irreplaceable loss.

The main premise of this guide is that no disaster plan can be transplanted intact from one campus, program, or even department, to another.

Records Management has not prepared a model plan that could be easily adapted for departmental use because such an approach violates the basic principles of sound records and information management. We are well aware there is no one single records classification system, retention schedule, or vital records program that is suitable for all departments, even within a single industry. A model plan approach would ignore the important role that local resources and conditions, such as available freezer space, fire protection, and weather patterns play in information disaster planning.

This guide presents you with some key presuppositions, concepts, elements, practical considerations, benefits, and steps that go into disaster planning. The application of the guide and wise use of its contents can provide the needed framework for a department to implement a comprehensive disaster prevention and recovery plan.

The importance of a written disaster recovery and vital records plan is also emphasized. The very process of writing the plan is an exercise in sound information management.

In-text Question 1

What is the first element in formulating a records disaster plan?

Answer

A clear understanding of disasters themselves - their types, causes, and effects.

2.3 Disaster Planning Requirements in Records Centres

A Disaster Plan guides your organisation through the proper responses to various types of disasters. This section highlights you some of the elements of a disaster plan.

1. Create a written disaster preparedness plan or policy, which includes disaster recovery, damage assessment, and post disaster evaluation procedures.
2. Identify and prioritize the most important records. This includes records needed to resume business, historical records, and collections. Determine which record media and collections are more vulnerable or valuable than others.
3. Analyse your building, site, and collection storage areas. Include building and site maps in your disaster plan.
4. Establish responses to all potential geographic and climatic hazards, and other risks which could jeopardize your employees, building, and collections: tornadoes; floods; fires, which will include water damage from fire hoses; pest infestation; mould; vandalism; and accidents.
5. Contact local civil defence offices to understand their disaster response procedures.
6. Identify sources of assistance, and develop contacts with appropriate consultants, suppliers, and vendors beforehand. Check your local Yellow Pages for contacts in your area, and make a list including names and telephone numbers. Update the list annually.
7. Establish contact with a freezer service; verify contact annually.
8. Special conservation efforts may be necessary with water or fire-damaged records, have phone numbers and addresses available of people or agencies to contact.
9. Include a copy of your collection inventory and vital records locator map in your disaster plan.
10. Include a supply list and locations in your disaster plan.

11. Create a telephone tree of staff and volunteers to help in the event of a disaster.
12. Establish a chain of command among staff members. All staff should know who they report to, and who they notify in case of disaster.
13. Know what your insurance carrier will require as evidence of damage: photographs, written documentation.
14. Establish salvage procedures for all collections, records, paper, and record media.

In-text Question 2
What is the first element of a disaster plan?

Answer
Creation of a written disaster preparedness plan or policy, which includes disaster recovery, damage assessment, and post disaster evaluation procedures

3.0 Study Session Summary and Conclusion

This session presented the basics for disaster plan, the need for the development of disaster plan and the disaster planning requirements in records centres.

4.0 Self-Assessment Questions

1. What are the basics for disaster plan?
2. What is the need for the development of disaster plan?
3. What are the disaster planning requirements in records centres?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2Sr9fso. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2T4jE2n and critique it in the discussion forum

6.0 References/Further Readings

Minnesota Historical Society.(2013). Disaster Preparedness. Retrieved from http://www.mnhs.org/preserve/records/disaster.php. Accessed on 24/1/2019.

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STUDY SESSION 3

Prevention Activity

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1 - Disaster Prevention Plan

2.2 - Basic Elements of Disaster Prevention Plan

2.3 - Disaster Prevention Methods in Records Centres

3.0Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos, Animations &Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome to study session 3, in this session, we shall be discussing disaster prevention activity, and this session will take you through the various aspect of prevention.

Disaster prevention is critical in the preservation of records in any environment. The prevention activity should be planned in order to achieve the aim. You as an individual or group who need to preserve their records effectively should plan the prevention methods to be used so that your records will stay for a long period of time.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Define disaster prevention plan.

2. Enumerate the basic elements of a disaster prevention plan.
3. Enumerate the disaster prevention methods in records centres.

2.0 Main Content

2.1 Disaster Prevention Plan

What do you understand by disaster **prevention plan?** You could explain this by saying, disaster prevention plan is a written, approved, implemented, and periodically tested program specifically outlining all actions to be taken to reduce the risk of avoidable disaster and minimise the loss should a disaster occur. The plan is based on a critical and thorough review of potential disasters to the facility, its geographic location or industry.

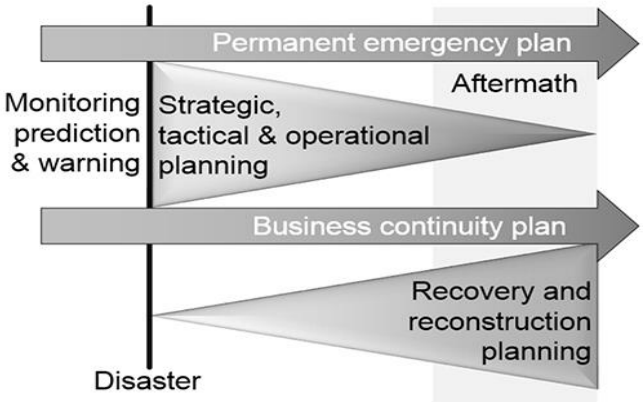

Fig 2.3.1: Disaster Prevention Plan

Your plan is a statement of appropriate responses for quick and cost-effective disaster recovery. It should identify which, where, and how stored records are to be protected and retrieved. Such reviews include the following issues:

1. Are microfilm, magnetic media and optical disks stored in appropriate environments?
2. Are smoking laws in full compliance in areas where paper or chemicals are stored?
3. Are boxes or other containers of records 12” to 14” off the floors (perhaps on pallets) even during initial processing activities?
4. Are chemicals, including those used by custodian and in office machines, stored in a manner to avoid or minimize disaster?
5. Are electrical appliances (i.e., coffee pots and other high heat producing items) turned off at the end of the day?
6. Should such appliances even be in an area used to house records?

7. Are proper security measures taken for the safety of the entire facility's contents?

In-text Question 1
What is a disaster prevention plan?

Answer
A disaster prevention plan is a written, approved, implemented, and periodically tested program specifically outlining all actions to be taken to reduce the risk of avoidable disaster and minimize the loss should a disaster occur.

2.2 Basic Elements of a Disaster Prevention Plan

In order for your disaster prevention plan to be effective, you should make it to contain certain basic elements. While the actual description of these elements will change from facility to facility, your experience shows that each should be included in the plan to achieve maximum effectiveness. Your key element of a disaster prevention plan includes, but is not limited to, the following:

1. Top-down directive for participation or buy-in
2. Clear policy statement
3. Establish activation authority
4. Task organization
5. Disaster recovery team
6. Facility floor plan or layout
7. Identification, location and protection of vital records and equipment
8. Information distribution procedures
9. Monitoring of destructive forces
10. Provisions for training of employees
11. Provisions for on-going review and revisions
12. Prearrangement for relocation and transportation provisions
13. Communications
14. Media relations
15. Up-to-date backups and hot or cold site locations

Disaster prevention is an essential part of good business. Every facility should have a prevention plan containing several key elements, including a clear policy statement, clearly defined activation authority, and a method to distribute information regarding the disaster. The plan should also provide for training employees and ongoing review and revisions of the written procedures.

2.3 Disaster Prevention Methods in Records Centres

The following are disaster prevention methods in records centres:

Fig 2.3.2: Disaster Prevention Methods in Records Centres

1. Make sure that your records are secured and only authorized access is granted.
2. Don't store items in direct contact with the floor. You should try to place boxes at least a few inches above the floor so as to minimise the threat of water damage.
3. Make sure your paper records are not stored in direct sunlight.
4. Insects can pose a problem for paper records so make sure to keep food and drinks away from the records. Make sure you have your storage area has frequent pest control checks.
5. For electronic records, make sure that there is a backup of the data on a regular basis.
6. Make sure your storage area has some kind of fire detection system.
7. Inspect your storage area on a regular basis for possible hazards or threats such as leaky water pipes, frayed electrical wires or other fire hazards.

In-text Question 2
What are some examples of basic elements of a disaster prevention plan?

Answer

Top-down directive for participation or buy-in, clear policy statement, disaster recovery team, facility floor plan or layout, provisions for training of employees etc.

3.0 Study Session Summary and Conclusion

This session presented the disaster prevention plan, the basic elements of a disaster prevention plan and the disaster prevention methods in records centres. They are presented in order of their arrangement.

4.0 Self-Assessment Questions

1. What is disaster prevention plan?
2. What are the basic elements of a disaster prevention plan?
3. What are the disaster prevention methods in records centres?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2GMQe2b. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2JCeQMg and critique it in the discussion forum

6.0 References/Further Readings

Prairie View A&M University. (2017). Records Disaster Recovery and Prevention. Retrieved from https://www.pvamu.edu/irm/records-management/records-disaster-recovery-prevention/. Accessed on 29/1/2019.

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STUDY SESSION 4

Response and Recovery Activities

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1 - Disaster Recovery Planning in Records Centres

2.2- Basic Elements of Disaster Recovery Plan in Records Cent

2.3 - Disaster Response Activities in Records Centres

3.0 Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos & Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome to study session 4, in the previous session, we discussed about disaster prevention plan this session will look at your response and recovery activity. Again you are welcome.

The response and recovery are the main activities in **disaster management** in records centres. They are the procedures through which the damaged records are managed. A lot of technical activities are involved here. They include identification of the type of disaster that occurred, sorting of the records that are not completely damaged, evacuation of the records among others.

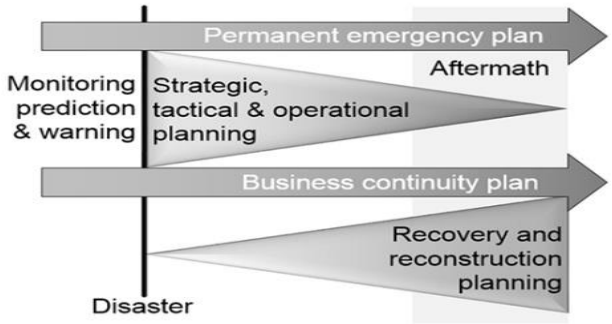

Fig 2.4.1 response and recovery

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the disaster recovery planning in records Centre's
2. Mention the basic elements of disaster recovery plan in records Centre's.
3. Explain the disaster response activities in records centres.

2.0 Main Content

2.1 Disaster Recovery Planning in Records Centres

Disaster recovery parallels disaster prevention. Like prevention planning, disaster recovery planning is good business. Disaster recovery is the process of resuming normal business operations, and reconstructing or salvaging vital and other important business records and equipment after a disaster.

An information disaster **recovery plan** is a written, approved, implemented, and periodically tested plan to reconstruct or salvage a department's vital and historical records and establishes procedures for the immediate resumption of business.

A comprehensive program that identifies and protects all vital records, particularly records stored electronically or on microforms, optical disk and CD generated records, swings the pendulum of post-disaster recovery from the salvage process to the reconstruction process. With such a program, the initial efforts centre around immediate relocation at a pre-established recovery site; the rapid reconstruction of essential records from computer tapes and disks, microfilm, and paper records; the prioritization of this reconstruction; and the actual resumption of normal business operations. In the absence of a comprehensive vital records program, the pendulum moves to the other extreme as the disaster team tries to identify and sort out which records are necessary to resume and continue business operations. The middle of a disaster is not the time to identify and sort records or determine their business resumption value.

In-text Question 1
What is disaster recovery in records centre?

Answer
Disaster recovery is the process of resuming normal business operations, and reconstructing or salvaging vital and other important records and equipment after a disaster.

2.2 Basic Elements of Disaster Recovery Plan

The following are the basic elements of a disaster recovery plan:

1. Clear policy statement, including recovery goals and objectives
2. Activation authority (WHO can activate the team's recovery plan)
3. Task organization (title and functions of each recovery team or team member)
4. Information distribution (specific methods for contacting recovery team members and alternates, vendors, support agencies, suppliers, and all those with whom special disaster contracts and agreements are in effect)
5. Red Tab Section, specific to disaster reactions (which disaster types will or will not be addressed in the plan)
6. Training of employees in recovery procedures
7. On-going review and revision of the plan

2.3 Disaster Response Activities

The following are the activities carried out during response to disaster in Records Centre's:

Temporary Off-Site Facilities

Should one or more temporary offsite facilities be needed during the emergency effort, the Emergency Response Team will look for available space. In the event of long-term records storage, other facilities or temporary

Fig 2.4.2: Temporary Off-Site Facilities

space may be required.

Supplies

While some supplies can be maintained at the facility to handle relatively small recovery efforts, keeping sufficient quantities of supplies to meet all contingencies is impractical, if not impossible. Emergency planning, however, should provide guidance on how and what supplies should be readily accessible.

Facility Integrity and Security

In the event of a major disaster at the facility, priority must be placed on facility security. Should there be any evidence of damage to the building structure (i.e., perimeter walls or roof) which would allow unauthorized access, immediate security precautions must be taken. The affected area should be cordoned off, and security personnel obtained to maintain 24-hour protection until building repairs can be made.

Treatment of Security Classified Materials

To prevent unauthorised disclosure of information or removal of records following a disaster, you should follow these steps: Head of records centre should be called to report to the vault. In the event of the Head of records centre absence, the assistant should be called. Wet records that can be salvaged by air drying should be moved to a secure location.

Treatment of Records Without Freezing

After the Emergency Response Team decides which material can be dried without freezing, drying rooms should be set up away from the affected area. If the air conditioning equipment and humidity controls used in the Archives stack area are still working, then drying of records should take place in this portion of the building. Relative humidity of 35-50% is optimum. Electric fans should be

used to circulate air on the documents. Work surfaces should be covered with plastic sheeting. Very little cleaning should be attempted on wet records that will not be frozen. After the documents are dry, muddy records can be brushed to remove the dirt. Attempting to remove mud while paper is wet forces the dirt deeper into the paper's fibres.

Treatment of Records to be Frozen

For stabilizing and restoring a large volume of water-damaged materials, freezing documents at low temperatures (below 20 degrees F) is the most effective method. Cold storage provides accessible and inexpensive space in which a large volume of material can be stabilized in the condition it was found, preventing further deterioration while awaiting treatment. It also provides time to assess the damaged material and to restore the building or stack area affected.

Treatment After Drying of Records

After the critical drying operation is over, you should return dry material and should be placed in the stack area and separated according to the different degrees of repair or restoration needed. Some documents may have escaped damage while others may require cleaning, flattening, or minor repairs.

Treatment of Photographic Materials and Microfilm

It is important to note that, Photographs, negatives, and microfilm stored in the Regional Archives are salvaged and restored in a different manner than are textual records and bound volumes. For emergency stabilization, wet black and white photographs, negatives, and microfilm should be sealed in polyethylene bags and placed in plastic (not metal) garbage cans under cold, clean running water. This should be done while the materials are still wet; they should never be allowed to dry before attempting to salvage. They may be left in running water for up to three days, although, treatment at a professional photo-finishing

laboratory equipped to handle damaged water photographs should begin as soon as possible.

Documentation of Disaster and Salvage Operations

In the event that a disaster does occur, a post-disaster assessment report should be written to determine the effectiveness of the recovery techniques utilized. Extensive photographs and written records of the conditions of the building and the procedures followed should be kept. It is also important to document all resources used to cope with a disaster, including personnel, materials, time, and expenses. This documentation can be important in helping to obtain emergency budgetary funds.

In-text Question 2
What are the examples of some of the disaster response activities?

Answer
Temporary off-site facilities, supplies, facility integrity and security, treatment of security classified materials, documentation of disaster and salvage operations etc.

3.0 Study Session Summary and Conclusion

This session presented the disaster recovery planning in records centres, the basic elements of disaster recovery plan in records centres and the disaster response activities in records centres.

5.0 Self-Assessment Questions

1. What is disaster recovery planning in records Centre's?
2. What are the basic elements of disaster recovery plan in records Centre's?
3. What are the disaster response activities in records centres?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2XsRsov. Watch the video & summarise in 1

paragraph

b. View the animation on add/sitehttps://bit.ly/2HNjiVp and critique it in the discussion forum

6.0 References/Further Readings

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STUDY MODULE 3

Digital Preservation and Techniques

Contents

Study Session 1: Strategies of Digital Preservation of Records

Study Session 2: The Need for Digital Preservation of Records

Study Session 3: Maintenance of Digital Records

Study Session 4: Requirements for a Digital Records Preservation

STUDY SESSION 1

Strategies of Digital Preservation of Records

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

- 2.1 - Planning, Pre-digitization, Digitization and Post- digitization Process
- 2.2 - Digital Preservation Infrastructure
- 2.3 - Digital Preservation Strategies

3.0 Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos, Animations &Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome to module 3, this session will discuss the concept of strategies of digital preservation of records.

Fig 2.4.2: Temporary Off-Site Facilities

We now live in a digital world, digital technology plays an important role in records preservation. Before adopting digital technology in records preservation, an individual (you) who is responsible for that needs to be very strategic. This is especially, if the desired result is to be achieved. The strategies that can lead to the achievement of this desired result include planning, pre-digitization, and digitization proper among others. Also, digital preservation infrastructure as well as approaches needed for the task is very important in the digital preservation activity.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Explain the steps of planning, pre-digitization, digitization and post-digitization process
2. Explain the digital preservation infrastructure
3. Discuss the digital preservation strategies

2.0 Main Content

2.1 Planning, Pre-digitization, Digitization and Post- digitization Process

1. Planning process
    a. Identification of an item to be digitized.
    b. Examining an item's legal rights.
    c. Evaluation of resources required.
    d. Finalise the standards to be followed
    e. Description of approaches and timing of quality mechanism.
    f. Evaluation of threats, hazard, comprising present and forthcoming shortcomings.

2. **Pre-digitization planning process**
    a. Selection of an item to be digitized.

b. Quality control of the materials to be digitized.
c. Materials to be digitized.
d. Any treatment required before digitization.
e. Collection of both descriptive and structural metadata.
f. Bibliographic and archival groundwork.

3. Digital conversion process

a. Digitization of selected materials.
b. Presence of professionally trained staff and equipment.
c. Quality regulator.
d. Formation of digital masters.

4. Post digitization process

a. Regulator of metadata connected to long-term preservation.
b. Compliance with information on distribution and repository structures, data gathering and administration.
c. Creation of digitized duplicates and metadata accessible online.
d. Valuation and assessment of the project.
e. Quality regulator.

In-text Question 1
What are the four digitization processes?

Answer
1. Planning, pre-digitization
2. Digitization and
3. Post digitization

2.2 Digital Preservation Infrastructure

Digital preservation will be achieved through a digital preservation infrastructure that: ensures data integrity, format and media sustainability, and information security. The following constitute digital preservation infrastructure:

a) Storage, network capacity, systems, and tools for the ingest or creation, processing, active file management, and preservation of born-digital files and digital surrogates.

(b) Processes to regularly review and update systems and tools that may be developed or procured to meet business needs.

(c) Affordable, managed, replicated content storage infrastructure for born-digital files and digital surrogates. Replication includes one preservation copy in a different storage environment, preferably in a remote geographic region, such as the replication that can be provided through cloud services.

(d) Tools to inventory all born-digital files and digital surrogates upon ingest.

(e) Tools for forensic identification and format characterization, which includes file format identification (identify the technical file types), format validation (confirm that the files meet documented format specifications), and technical metadata extraction (documenting how the files were created, including the applications and operating systems) which is used to support policy-based assessment of format obsolescence risks and to present the file to users using the appropriate application or viewer in context.

(f) Tools for file format transformations to perform file migrations over time as formats become obsolete and at-risk.

(g) Standardized workflow processes for associating born-digital and digital surrogate files with record identifiers and metadata and ensuring that files are in appropriate preservation storage and access server locations (on premise or in the cloud).

2.3 Digital Preservation Strategies

You have several approaches, some more practical than others, to ensure that electronic records remain useful over time. One is to save all of the hardware, software, and documentation needed to support the records. Known as the

"**computer museum**" approach, it is not very realistic on a large scale because, given how rapidly hardware and software environments change, it means storing and maintaining huge quantities of out-dated equipment with no assurance that any of it will work when needed.

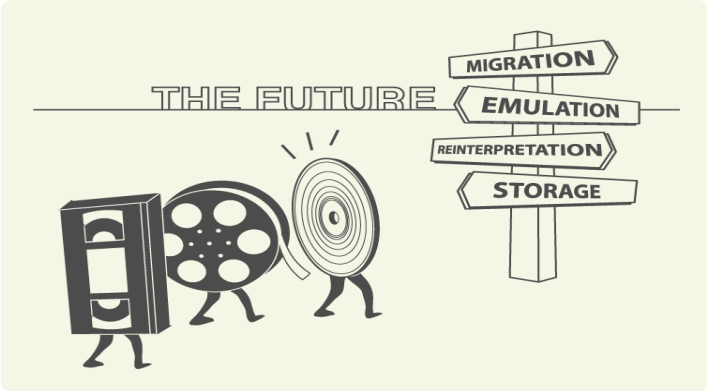

Fig 3.1.2: Digital Preservation Strategies

Emulation has a similarly antiquarian flavour. Emulator programs simulate the behaviour, look, and feel of other programs, thus preserving the functionality of the records in their original format without the necessity of saving the original equipment and software. However, emulation has so far proven more attractive in theory than in practice.

There are few examples of success using this approach, and costs have proven high. It has a further limitation in that, at best, emulation simply reproduces earlier, less sophisticated versions of an application. Given all the expenses of technology, it seems problematic to limit the value of information by preserving it in a static framework.

Encapsulation is a third approach to preservation. It involves combining the object to be preserved with all of the necessary details of how to interpret it within a wrapper or package, all possibly formatted in XML. While appealing in its comprehensiveness, encapsulation has several drawbacks: file sizes are large because of all of the included information; format specifications must be determined; the encapsulated records must somehow be generated, usually separate from the act of record creation; and the encapsulated records must still be migrated over time.

The most common approach to preserving electronic records involves a combination of two other techniques: migration and conversion. You can describe migration as the process of moving files to new media (also known as "refreshing") or computer platforms in order to maintain their value. On the other hand, conversion entails you changing files from one format from one to another and may involve moving from a proprietary format, such as Microsoft Word, to a non-proprietary one such as a plain text file or XML.

To avoid losing data in the process, you should perform initial tests and analysis to determine exactly what changes will occur and whether they are acceptable. With both migration and conversion, special attention must be paid to also maintaining the accessibility of any associated metadata. When properly planned and executed, the migration and conversion approach probably represents the easiest and most cost-effective preservation method available today.

In-text Question 2
What are the most common approaches to preserving electronic records?

Answer
1. Migration and
2. Conversion.

3.0 Study Session Summary and Conclusion

This session presented the steps of planning, pre-digitization, digitization and post-digitization Process, the digital preservation infrastructure and the digital preservation strategies.

4.0 Self-Assessment Questions

1. What are the steps of planning, pre-digitization, digitization and post-digitization process?
2. What is digital preservation infrastructure?

3. What is migration?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2tEVix6. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2W4nOc9 and critique it in the discussion forum

6.0 References/Further Readings

Doyle, J., Viktor, H. and Eric, P. (2009). Long-term digital preservation: preserving authenticity and usability of 3-D data. *International Journal on Digital Libraries*, 10 (1), 33–47. Retrieved from https://link.springer.com/article/10.1007/s00799-009-0051-7. Accessed on 24/1/2019.

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STUDY SESSION 2

The Need for Digital Preservation of Records

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content essence

2.1 - Digital Continuity

2.2- Importance of Digital Preservation

3.0 Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos &Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome to study session 2, this session will explore digital preservation of records.

Your **digital preservation** needs to be maintained in order to achieve digital continuity. This is because some records needs to be preserved for a long period of time. **Digital archiving** is used to preserve digital records in order to achieve that. Databases are used to preserve the digital records for future use. Hence, achieving digital continuity needs continuous maintenance of digital records.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the digital continuity.
2. List the importance of digital preservation.

2.0 Main Content

2.1 Digital Continuity

An important aspect of digital continuity is digital preservation. Many records have retention periods longer than the life of the systems they are captured in. Digital preservation requires a proactive program to identify records at risk and take necessary action to ensure their on-going viability.

Preserving digital and audio-visual information and records may require migration to new platforms and formats. Regular and planned migration helps avoid obsolescence and ensures information continues to be accessible and usable.

Digital records are subject to three potential forms of obsolescence:

1. The physical carrier of the record becomes obsolete - Standard media of the 1980s, such as 8 inch and 5¼ inch floppy disks, are no longer commercially available. Over time, current media such as CDs and DVDs will also become obsolete.
2. The hardware needed to access the record becomes obsolete - Both the drives needed to read the media and the computers required to operate them have become obsolete in our rapidly changing world. Most new computers have a usable life of only three to five years and are not fitted with floppy disk drives.
3. The software needed to access the record becomes obsolete - This includes either the software to read and write the record, the operating system to run the software, or both.

In-text Question 1
What are the potential forms of obsolescence to digital records?

Answer
1. Physical carrier,
2. Hardware and
3. Software.

2.2 Importance of Digital Preservation

Your main purpose of digital preservation is to ensure protection of information of enduring value for access by present and future generations. Digital technology is revolutionising the traditional concepts of preservation, access and archival of information. Some important purposes which fulfil the aims of digital preservation are:

Fig 3.2.1: Digital Preservation Strategies

1. For maintaining the historical value of information
2. For reducing the effects of deteriorating factors
3. For saving the space and time in all respect.
4. For making information survive life time
5. For providing worldwide accessibility
6. For the purpose of backup

In-text Question 2
Itemize some of the importance of digital preservation

Answer
1. for maintaining the historical value of information,
2. for reducing the effects of deteriorating factors,
3. for saving the space and time in all respect etc.

3.0 Study Session Summary and Conclusion

This session presented the digital continuity and the importance of digital preservation. They are presented in order of their arrangement in this session.

4.0 Self-Assessment Questions

1. What is digital continuity?
2. What is the importance's of digital preservation?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2XsUxoz. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2EubDKD and critique it in the discussion forum

6.0 References/Further Readings

Anil, K. J. (2018).Importance of Digital Preservation in Digital Era. Retrieved From https://www.researchgate.net/publication/326534604_IMPORTANCE_OF_DIGITAL_PRESERVATION_IN_DIGITAL_ERA.Accessed on 24/1/2019.

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STUDY SESSION 3

Maintenance of Digital Records

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

- 2.1 - Purpose of Maintaining Digital Records
- 2.2 - Sources of Digital Records
- 2.3 - Storage and Maintenance of Digital Records

3.0 Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos &Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome to study session 3, maintenance of digital records will be our focus of interest in this session. We shall look at various aspect of maintenance of digital records.

Digital records are created for their usage. Some of them need to be kept for a long period of time. As such, they are maintained in order to achieve that. This is because without maintaining them, you cannot be used in the future when the need arises.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the purpose of maintaining digital records.
2. Explain the sources of digital records
3. Explain the storage and maintenance of digital records

2.0 Main Content

2.1 Purpose of Maintaining Digital Records

For you to have a long-term digital preservation, the process of maintaining digital objects through time to ensure continued access, has become a crucial issue in recent years. Whilst the amount of digitised information is constantly increasing, so too is the pace of progress in information technology, resulting in obsolescence of the software and hardware required to access and view digital information. Despite many organisations recognising this threat and the resulting need for preservation action, more work is required to effectively address the issue.

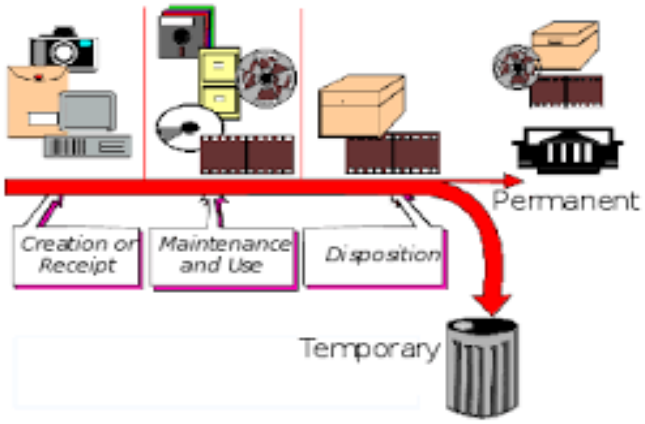

Fig 3.3.1: Purpose of Maintaining Digital Records

You should note that, to ensure that active digital records are readable for future use, it is recommended that you have a **proactive maintenance** plan be implemented. Your maintenance plan would involve monitoring and migrating records when operating systems and/or software applications are changed or upgraded. It is best practice you create a documentation trail when files are migrated from one system to another. Documentation should include: systems and software applications, date of migration, name, and job title of person responsible for migration, and description of any loss of information that may occur during the migration process.

2.2 Sources of Digital Records

1. **Electronic documents**: Electronic documents from authoring tools will be captured in the RMA. These documents will come into the RMA along with the associated metadata. Once electronic documents are declared

records, the associated metadata will transfer to the records management application, while the content of the document will remain in the document repository.

2. **Imported electronic documents**: The system will also provide for the capture of existing electronic documents that may be received from outside sources. During the input process, metadata will be captured along with the content of the document.
3. **Scanned paper documents**: Hard copy paper documents will be scanned into the system and the associated metadata will be manually entered. Once the scanned documents are in the system, they will behave as electronic documents in that their metadata will be transferred to the RMA, while the image itself will remain in the document repository.

In-text Question 1
What is long-term digital preservation?

Answer
Long-term digital preservation is the process of maintaining digital objects through time to ensure continued access.

2.3Storage and Maintenance of Digital Records

As part of a records management plan for electronic records, you will need to determine where and how your records will be stored. This decision will be based on the likelihood of access of those resources versus the overall cost in maintaining them. Your general options for storage include online, near-line, and offline. Records that are accessed most should be stored online; records accessed the least offline. After it is determined where the records will be stored you will need to determine how to provide various users with access. This will depend on the records themselves as well as the type of media chosen. You should select appropriate media and systems for maintaining your records for the required period of time. Regardless of your choice, files may need to be refreshed (transferred to new media) or migrated (to a different format) possibly

within a pre-determined period of time. For more information about the types of digital media please review the *Digital Media* section of these guidelines. Online, near-line, and offline storage are described below.

Online Storage

Online storage allows immediate access to records to anyone on the system's network. Properly designed storage will provide access to appropriate users only. Online storage maintains the greatest functionality but requires more expensive network storage.

Examples of online storage include:

1. Storage Area Networks (SAN). SANs allow access to remote drives with the same convenience of internal hard drives. A SAN is a networked system.
2. Just a Bunch of Disks (JBOD). JBOD is a collection of disks that are set up to look like one large disk to users. If one disk goes bad, only the information on that disk is lost; the information on the others is still readable. This is sometimes used to store backup files.
3. Redundant Array of Inexpensive/Independent Disks (RAID). RAID uses a group of disks to back up data on a daily basis. There are many different configurations of RAID arrays with their own features. RAID is frequently used, but depending on the size of your institution, may or may not be cost effective. Although RAID is used to backup local files, it is not intended to be used as a main backup method.
4. Cloud Computing. More recently, online storage also refers to files that can be accessed online via the Internet. With cloud computing files are stored by a third-party and accessed through a web service. This technology continues to be explored. It may not be appropriate for confidential files or vital records that need to have 24 hour access. As

with any outside provider, questions need to be asked about their policies and procedures for storing, preserving, and providing access to records.

Near-line Storage

Near-line storage is storage in a system that is not a direct part of your network, but that can be accessed through your network (e.g., an optical media jukebox). Access to these files is done with an automated process that selects the correct disk/tape from a disk/tape library and makes it accessible. This option maintains a moderate amount of functionality. While the storage space is cheaper than online storage, near-line storage requires extra time to manipulate both the files and media of choice to access the records. Near-line storage is often used for backups as large quantities of data can be managed quickly.

Offline Storage

Files that cannot be accessed immediately are said to be stored offline (e.g., files not accessible through your network such as on removable media like external hard drives or magnetic tape). Older records or records that do not need to be accessed frequently are often stored online. This option trades functionality for stability. There is a focus on data accuracy, protection, and security due to the long-term storage necessity. The longer records need to be maintained the more important preservation methods, back-up procedures, storage conditions, handling procedures, and security become. Offline storage can be stored in-house, off-site, or outsourced.

Examples of offline storage include:

1. Removable magnetic or optical media (tape, DVD). For integrity purposes, read-only media is preferred.
2. Flash media (solid state media has no moving parts)
3. External hard drives (with moving parts)

In-text Question 2
What are the examples of online storage?

Answer
1. Storage Area Networks (SAN),
2. Just a Bunch of Disks (JBOD),
3. Redundant Array of Inexpensive/Independent Disks (RAID) and
4. Cloud computing.

3.0 Study Session Summary and Conclusion

This session presents the purpose of maintaining digital records, the sources of digital records as well as storage and maintenance of digital records. They are presented in order of their arrangement in the session.

4.0 Self-Assessment Questions

1. What is the purpose of maintaining digital records?
2. What are the sources of digital records?
3. What are storage and maintenance of digital records?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2GRL3hB. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2wfbeaD and critique it in the discussion forum

6.0 References/Further Readings

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Minnesota Historical Society.(2012). Digital Media Storage. Retrieved from http://www.mnhs.org/preserve/records/electronicrecords/erstorage.php. Accessed on 24/1/2019.

The Regents of the University of Michigan. (2017). Identification, Maintenance, and Preservation of Digital Records Created by University of Michigan. Retrieved from http://www.spg.umich.edu/policy/601.08-1.Accessed on 24/1/2019.

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STUDY SESSION 4

Requirements for a Digital Records Preservation

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

- 2.1 - Requirement Needed Before Developing Digital Preservation Plan
- 2.2 - Physical Storage Options
- 2.3 - File Formats
- 2.4 - Digital Preservation Techniques

3.0 Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos &Out of Class activities)

6.0 References/Further Readings

Introduction

You are welcome to study session 4, this will be our last discussion for the course, in this session, will explore the concept of digital records preservation.

Digital preservation cannot be carried out without requirements. Some of these requirements are identification of a particular file format, storage technique, preservation strategy etc. All these are needed in order to achieve the desired result.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the requirement needed before developing digital preservation plan.
2. Explain physical storage options.

3. Explain file formats.
4. Discuss digital preservation techniques

2.0 Main Content

2.1 Requirement Needed Before Developing Digital Preservation Plan

As a first step in developing your preservation plan, you should do a need assessment to help guide your decisions. While the complexity of such an analysis will vary from situation to situation, these basic components should always be included.

Fig 3.4.1: Requirement Needed Before Developing Digital Preservation Plan

First, you need to understand the value of your information. The value of your information will justify your investment in technology, over the short- and the long-term. Minnesota's enterprise technical architecture notes that information is the state's most important asset. But all information is not created equal; some has much more value than others. Some of your information, as records, will have legal and evidentiary significance and may well demand special attention. Most of the information you want to preserve will be important to your agency's mission or, increasingly, to the business of other agencies as well. As e-government develops in complexity and sophistication, more and more agencies will be expected to work within the framework of a common technological architecture and to share the information they create.

The practical side of understanding the value of your records is determining their retention requirements. How long do you really need to keep them? Why are you keeping them? Do they have to be kept in electronic format or is there

another, more cost-effective option for long-term storage? For instance, a word processing document might be printed and kept as a paper record without losing any of its value. In contrast, printing a web page means a significant loss of information and functionality.

It is also important for you to ascertain if access to certain data in your records is restricted by statute. The state's Government Data Practices Act and some federal laws, such as the Health Insurance Portability and Accountability Act (HIPAA), will determine if data needs to be protected as confidential or non-public. If it does, then you will need to ensure that your long-term storage and access policies account for those obligations.

2.2 Physical Storage Options

As mentioned above, choosing the most appropriate storage option for your situation will depend upon your records' access requirements. There are basically three options available to you:

Fig 3.4.2: Physical Storage Options

1. Online storage: Records are kept on a server or hard drive and are immediately available for use over a network. This option is best for records that must be accessed frequently.
2. Near-line storage: Records are stored on media such as optical disks in jukeboxes or tapes in automated libraries which are attached to a network. Because retrieval is slower than with online storage, this option is most appropriate for records that are accessed occasionally.

3. Offline storage: Records are stored on removable media and must be manually retrieved. This option provides the slowest access and should be used for records that are only rarely needed.

In-text Question 1
Itemize the physical storage options?

Answer
1. Online storage,
2. Near-line storage and
3. Offline storage

2.3 Required File Formats

1. **Audio** — Highly recommended: AIFF (.aif, .aiff) or WAV (.wav). Moderately recommended: MP3 (.mp3), AAC (.mp4, .mp4a, .aac), FLAC (.flac) or ALAC (.m4a).
2. **Video** — Highly recommended: Uncompressed Quick time Movie (.mov); uncompressed AVI (.avi). Moderately recommended: MPEG-1, MPEG-2 or MPEG-4 encoded video (.avi, .mpg, .mpeg, .mov, .mkv, .mp4).
3. **Virtual Reality/3D** — Highly recommended: X3D (*.x3d). Please talk with a North-western Digital Librarian to determine best output settings and format.
4. **Image** - Highly recommended: Full colour images @ 600dpi or higher saved as JPEG2000 Lossless (.jp2) or TIFF 24-bit, uncompressed (.tif, .tiff). Moderately recommended: lossy compressed formats limited to JPEG (.jpg, .jpeg), JPEG2000 (.jp2), TIFF (.tif, .tiff), or PNG (.png) at highest quality possible.
5. **Text** - Highly recommended: Open Document Text (.odt), UTF-8 Unicode text (.txt), or PDF/A (.pdf). Moderately recommended: Rich Text Format (.rtf).

6. **Presentation** - Highly recommended: Open Document Presentation (.odp). Moderately recommended: PDF/A (.pdf) for images only.
7. **Spread sheet** - Highly recommended: Open Document Spreadsheet (.ods). Moderately recommended: Comma separated value CSV (.csv) or Tab-delimited text file (.txt).

2.4 Digital Preservation Techniques

You have several approaches, some more practical than others, to ensure that electronic records remain useful over time. One is to save all of the hardware, software, and documentation needed to support the records. Known as the "computer museum" approach, it is not very realistic on a large scale because, given how rapidly hardware and software environments change, it means storing and maintaining huge quantities of outdated equipment with no assurance that any of it will work when needed.

Fig 4.4.3: Digital Preservation Techniques

Emulation has a similarly antiquarian flavour. Emulator programs simulate the behaviour, look, and feel of other programs, thus preserving the functionality of the records in their original format without the necessity of saving the original equipment and software. However, emulation has so far proven more attractive in theory than in practice. There are few examples of success using this approach, and costs have proven high. It has a further limitation in that, at best, emulation simply reproduces earlier, less sophisticated versions of an

application. Given all the expenses of technology, it seems problematic to limit the value of information by preserving it in a static framework.

Encapsulation is a third approach to preservation. It involves combining the object to be preserved with all of the necessary details of how to interpret it within a wrapper or package, all possibly formatted in XML. While appealing in its comprehensiveness, encapsulation has several drawbacks: file sizes are large because of all of the included information; format specifications must be determined; the encapsulated records must somehow be generated, usually separate from the act of record creation; and the encapsulated records must still be migrated over time.

The most common approach to preserving electronic records involves a combination of two other techniques: migration and conversion. Migration is the process of moving files to new media (also known as "refreshing") or computer platforms in order to maintain their value. Conversion entails changing files from one format from one to another and may involve moving from a proprietary format, such as Microsoft Word, to a non-proprietary one such as a plain text file or XML. To avoid losing data in the process, you should perform initial tests and analysis to determine exactly what changes will occur and whether they are acceptable. With both migration and conversion, special attention must be paid to also maintaining the accessibility of any associated metadata.

When properly planned and executed, the migration and conversion approach probably represents the easiest and most cost-effective preservation method available today.

In-text Question 2
What are some examples of file formats?

Answer
PDF, ODP, ODT, ODS, JPEG etc.

3.0 Study Session Summary and Conclusion

This session presented the requirement needed before developing digital preservation plan, the physical storage options, the required file formats as well as digital preservation techniques.

4.0 Self-Assessment Questions

1. What is the requirement needed before developing digital preservation plan?
2. What are physical storage options?
3. What are the examples of file formats?
4. What are digital preservation techniques?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube addhttps://bit.ly/2tGhsid. Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/30Ewg0e and critique it in the discussion forum

6.0 References/Further Readings

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Northwestern University.(2019). File Format Recommendations. Retrieved fromhttps://www.library.northwestern.edu/about/administration/policies/file-format-recommendations.html. Accessed on 24/1/2019.

Rouse, M. (2019). File Format. Retrieved from https://whatis.techtarget.com/definition/file-format. Accessed on 29/1/2019.

Surendra, K. P. (2019). Digital Preservation Methods and Technology In Digital Era. Retrieved from http://www.academia.edu/13486465/Digital_Preservation_Methods_and_Technology_In_Digital_Era. Accessed on 29/1/2019.

XIII Glossary

active record

A record needed to perform current operations or ongoing business matters. It is consulted frequently, and it must be conveniently available for immediate reference, either manually or via a computer system.

administrative retention criteria

The standards or rules concerned with the availability of records for long-term administrative consistency and continuity, as well as for day-to-day operations of individual program units.

admissibility

A legal term that refers to the conditions under which evidence, including records and information, may be introduced into legal proceedings. To be admissible as evidence, a record must satisfy two basic requirements: (1) Its content must be relevant to the issue at hand; and (2) its authenticity must be firmly established.

archival authority

Agency or program responsible for selecting and preserving archives, making them available, and approving destruction of other records.

archival data

Information typically stored on secondary, off-line media and is not directly accessible to users of a computer system, but it is nevertheless required for retention.

archives

Documents created or received and accumulated by a person or organization in the course of the conduct of affairs and preserved because of their continuing value; the building or part of the building in which archives are preserved and made available for consultation, and/or the agency or program responsible for selecting, acquiring, preserving, and making available archives.

audit

Independent review and examination of records and activities to test for compliance with established policies or standards, often with recommendations for changes in controls or procedures.

authenticity (of a record)

Efforts to establish that a record is what it purports to be and has not been tampered with or otherwise corrupted since its creation; that is, its reliability can be demonstrated through the various phases of its lifecycle.

certificate of destruction

A form used to document the destruction of specified records series based on the records retention schedule. Typically includes the records series title, dates covered in the series, organization/agency unit owner of the records series, volume or number of boxes destroyed, name of destruction authorizer, method of destruction, and date of destruction.

confidential record

Any business record containing sensitive information. Such records are typically subject to special rules and policies concerning their access, use, and disclosure.

convenience copies

Duplicate(s) of an original record, often used for access or reference.

data archiving

The process of transferring electronic records from online storage devices, such as hard drives, to removable recording media, such as magnetic tape or optical disks, for offline storage. In most cases, the archived data is relatively inactive and does not need to be accessible online.

data migration

The process of moving data from one information system or storage medium to another; the process of converting electronic records to new file formats or storage media to maintain their usability over time.

data warehouse

A large database structure to support decision-making.

database

A file containing records organized into one or more data elements, called fields, which store particular categories of information.

destruction

The definitive obliteration of a record beyond any possible reconstruction. Methods may include shredding or incinerating paper records, or performing secure data deletion or disk and memory wiping for electronic records. See datamgmt.iu.edu for guidance on the destruction of institutional data.

destruction hold

A hold placed on the scheduled destruction of records due to foreseeable or pending litigation, governmental investigation, audit, or special organizational requirements. Also called legal hold, freeze notice, preservation order, suspension order, or hold notice.

digital document images

Images that are true copies of the documents from which they were made. A true copy is one that accurately reproduces an original document.

digital document imaging systems

Bit-mapped image representation systems that can complement, supplement, or replace paper filing systems and microfilm.

directory

A table of contents for an electronic storage medium.

disposition

A final administrative action taken with regard to records, including destruction, transfer to another entity, or permanent preservation. Examples may include destruction (see definition listed above), transfer to the University Archives, or to a centralized office.

document management

Techniques used to regulate the creation, use, and maintenance of documents according to established policies and procedures. See related: enterprise content management.

document type definition (DTD)

A set of rules that specify the structure of a document and the tags used to define that structure and that can be used to validate whether a document is well formed.

electronic record

A record that contains machine-readable, as opposed to human-readable, information.

electronic signature

Any electronic method of signing a computer-processable record.

enterprise content management

The technologies, tools, and methods used to create, capture, process, store, deliver, and preserve information content, particularly unstructured content, across an enterprise. See related: document management

event-based disposition

A concept wherein once a registered event occurs, the disposition schedule starts. Related: trigger event.

event code

A unique identifier of a specific event which starts the disposition period; often an abbreviation of a phrase or word. Example: the code for “calendar year end” may be CYE.

FERPA

Family Educational Rights and Privacy Act (20 U.S.C. § 1232g; 34 CFR Part 99). A federal law that protects the privacy of student education records. The law applies to all schools that receive funds under an applicable program of the U.S. Department of Education.

field

A data element within a database record.

file plan

A classification scheme describing different types of files maintained in an office, how they are identified, where they should be stored, and how they should be indexed for retrieval, and a reference to the approved disposition for each file.

general records schedule

A records schedule governing specified series of record common to several or all agencies or administrative units of a corporate body, which are sometimes characterized as functional retention schedules.

Generally Accepted Recordkeeping Principles (GARP)

Guiding statement on information governance based on eight recordkeeping principles published by ARMA International: accountability, integrity, protection, compliance, availability, retention, disposition, and transparency.

HIPAA

Health Insurance Portability and Accountability Act of 1996. A set of national standards for the protection of certain health information.

historical record

Material in any format (paper, electronic, audiovisual, etc.) created or collected by an Indiana University unit or representative that is preserved because of the enduring administrative, legal, fiscal, evidential, or historical value or as evidence of the functions and responsibilities of the unit or individual."

inactive record

A record no longer needed to conduct current business but preserved until it meets the end of its retention period.

information life cycle

The distinct phases of the existence of information, from creation to final disposition.

key field

A field selected for indexing within records contained in a database.

metadata

Structured information that describes, explains, locates, or otherwise makes it easier to retrieve, use, or manage an information resource.

migration

The process of moving data from one information system or storage medium to another. Note: Migration is done to ensure continued access to the information as the system or medium is replaced, becomes obsolete, or degrades over time.

office of record

An office designated to maintain the record or official copy of a particular record in an organization.

office retention

The period of time that a records series is to be maintained in active storage for regular use.

offline information

Information stored apart from the device on which it will be retrieved or played back.

online information

Information immediately and continuously available to a computer.

permanent record

A record that has been determined to have sufficient historical, administrative, legal, fiscal, or other value to warrant continuing preservation.

physical inventory

The process of identifying all tangible records of an organization.

preservation

The process and operation involved in ensuring the technical and intellectual survival of authentic records through time.

privileged record

A record that contains information that is accessible only to those authorized to view it.

real-time backup

A disaster recovery method that employs simultaneous or nearly simultaneous recording of information by multiple computers.

record(s)

Information in any form created, received, or maintained by Indiana University or its employees or agents pursuant to legal obligations or in pursuit of the University's mission.

record copy

The copy of a record designated to satisfy an organization's retention requirements for information that exists in multiple copies; also known as the official copy.

record series

A group of logically related records that support a specific business or administrative activity, e.g., a personnel file consisting of an application, reference letters, benefit forms, etc.

record series code

A unique identifier for a record series; often comprised of alphanumeric digits.

records inventory

A detailed listing that could include the types, locations, dates, volumes, equipment, classification systems, and usage data of an organization's records; the dissecting of each record to capture all pertinent information about the record to be used in its appraisal.

records management

The field of management responsible for the efficient and systematic control of the creation, receipt, maintenance, use and disposition of records, including processes for capturing and maintaining evidence of and information about business activities in the form of records.

records retention schedule

A list of records series maintained by all or part of an organization, together with the period of time that each series is to be kept.

records survey

A broad overview of the quantity and type of records within an organization, which is usually completed prior to a detailed inventory.

retention

An aspect of records management that determines how long records need to be kept.

retention period

The length of time a record must be kept to meet administrative, fiscal, legal, or historical requirements.

retention schedule

A comprehensive list of records series, indicating for each the length of time it is to be retained and its disposition.

security copies

Additional record copies created as protective measure; also known as backup copies.

source documents

Paper documents that contain information to be converted to electronic records.

student record

Records that are directly related to a student and maintained by an educational agency or institution or by a part acing for an agency or institution, if certain conditions are met.

tagging

A specific piece of metadata- keyword term or phrase- that is assigned to a content object for use in browsing or searching. The process of tagging generally refers to the ability of users to collaboratively add their own descriptive tags to information.

total retention

The amount of time a records series is to be stored in active and inactive storage.

trigger event

Something that must occur before a retention period begins to run. Related: event-based disposition.

unit-specific retention schedule

A unit-specific retention schedule prepared specifically for a unit in an organization. It lists only those record series that a given unit maintains with unequivocal retention designations for each.

vital records

Records that are fundamental to the functioning of an organization and necessary to continue operations without delay under abnormal circumstances.

vital records program

A set of policies and procedures for the systematic, comprehensive, and economical control of losses associated with vital records.